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Watson et al.

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(45) **Date of Patent: Sep. 2, 2025**

(54) **DEVICE FOR ADJUSTING A SEAT POSITION OF A BICYCLE SEAT**

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(72) Inventors: **David Watson**, North Vancouver (CA); **Saman Namvar**, West Vancouver (CA)

(73) Assignee: **David Watson**, North Vancouver (CA)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 233 days.

(21) Appl. No.: **17/923,201**

(22) PCT Filed: **May 4, 2021**

(86) PCT No.: **PCT/CA2021/050623**

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(2) Date: **Nov. 3, 2022**

(87) PCT Pub. No.: **WO2021/223022**

PCT Pub. Date: **Nov. 11, 2021**

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Related U.S. Application Data

(60) Provisional application No. 63/019,626, filed on May 4, 2020.

(51) **Int. Cl.**
B62J 1/08 (2006.01)
B62J 43/28 (2020.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B62J 1/08*** (2013.01); ***B62J 45/20*** (2020.02); ***B62J 45/415*** (2020.02); ***B62J 45/423*** (2020.02);
(Continued)

(58) **Field of Classification Search**
CPC ... B62J 1/08; B62J 45/20; B62J 45/413; B62J 45/414; B62J 45/4151; B62J 45/4152; B62J 43/28; B62J 2001/085; G01C 9/06
See application file for complete search history.

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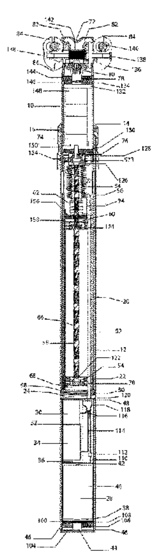
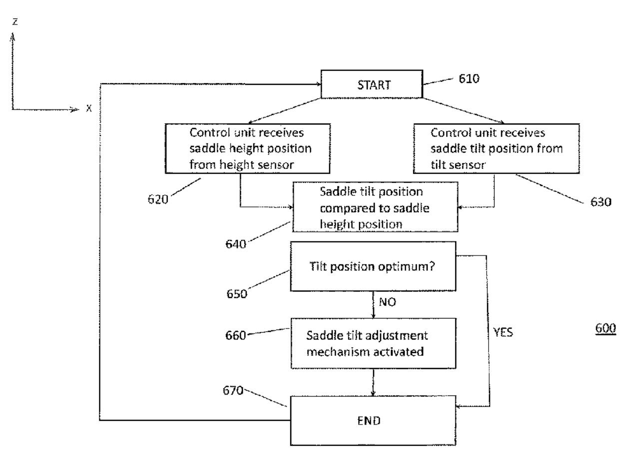
Primary Examiner — Huan Le
(74) *Attorney, Agent, or Firm* — Davis Wright Tremaine LLP; Heather M. Colburn

(57) **ABSTRACT**

There is described a device for adjusting a tilt of a bicycle seat. An angle of a seat coupling relative to a seat post coupling is determined. An angle of the seat post coupling relative to a reference position, such as a horizon, is determined. For example, the angle of the seat post coupling relative to the reference position may be determined based on a current geographic position of the bicycle. Based on the determined angles, the tilt of the bicycle seat is adjusted by actuating a seat tilt driver, such as a motor.

24 Claims, 26 Drawing Sheets





(51) **Int. Cl.**

B62J 45/20	(2020.01)
B62J 45/415	(2020.01)
B62J 45/423	(2020.01)
G01C 9/06	(2006.01)

(52) **U.S. Cl.**

CPC ***G01C 9/06*** (2013.01); *B62J 2001/085* (2013.01); *B62J 43/28* (2020.02)

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FIGURE 1

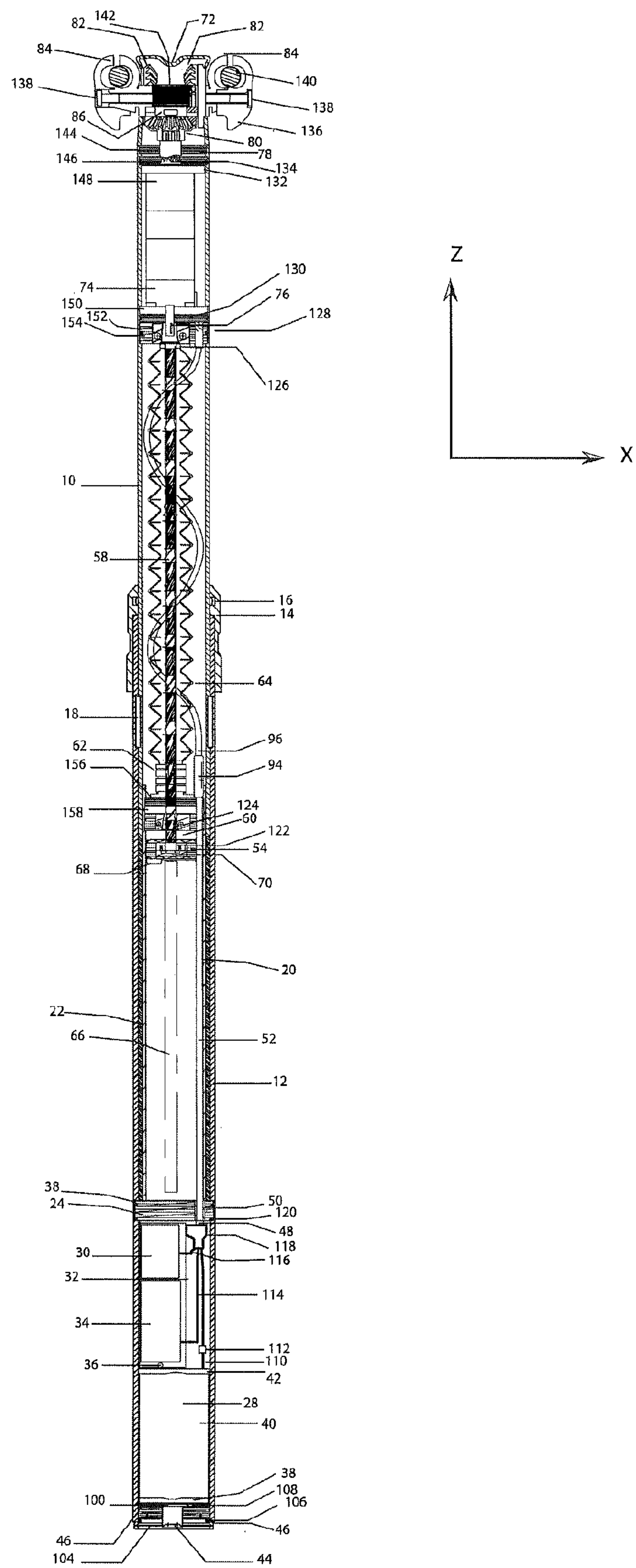
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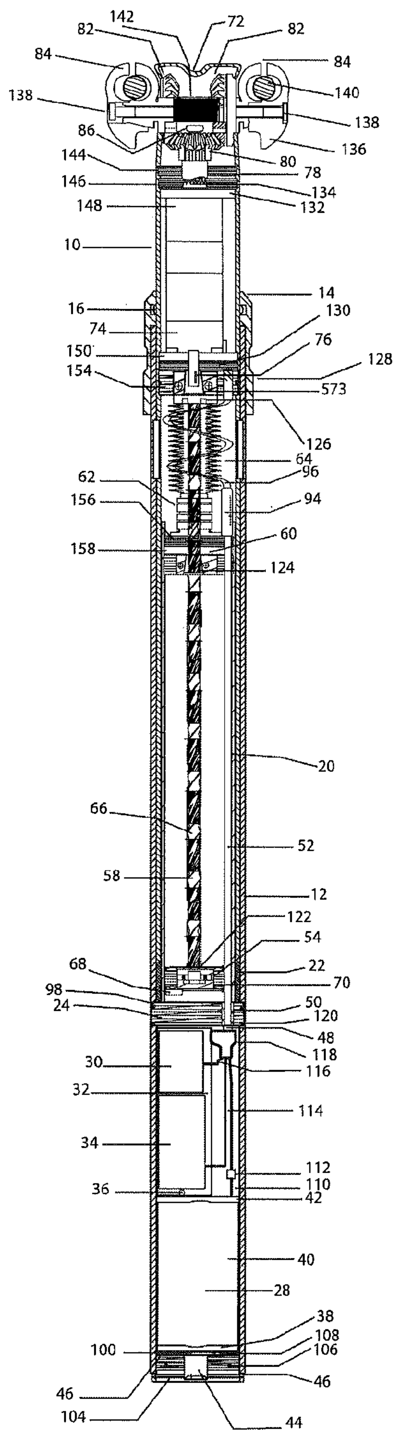
FIGURE 2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573
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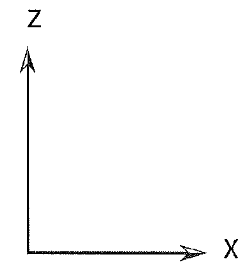
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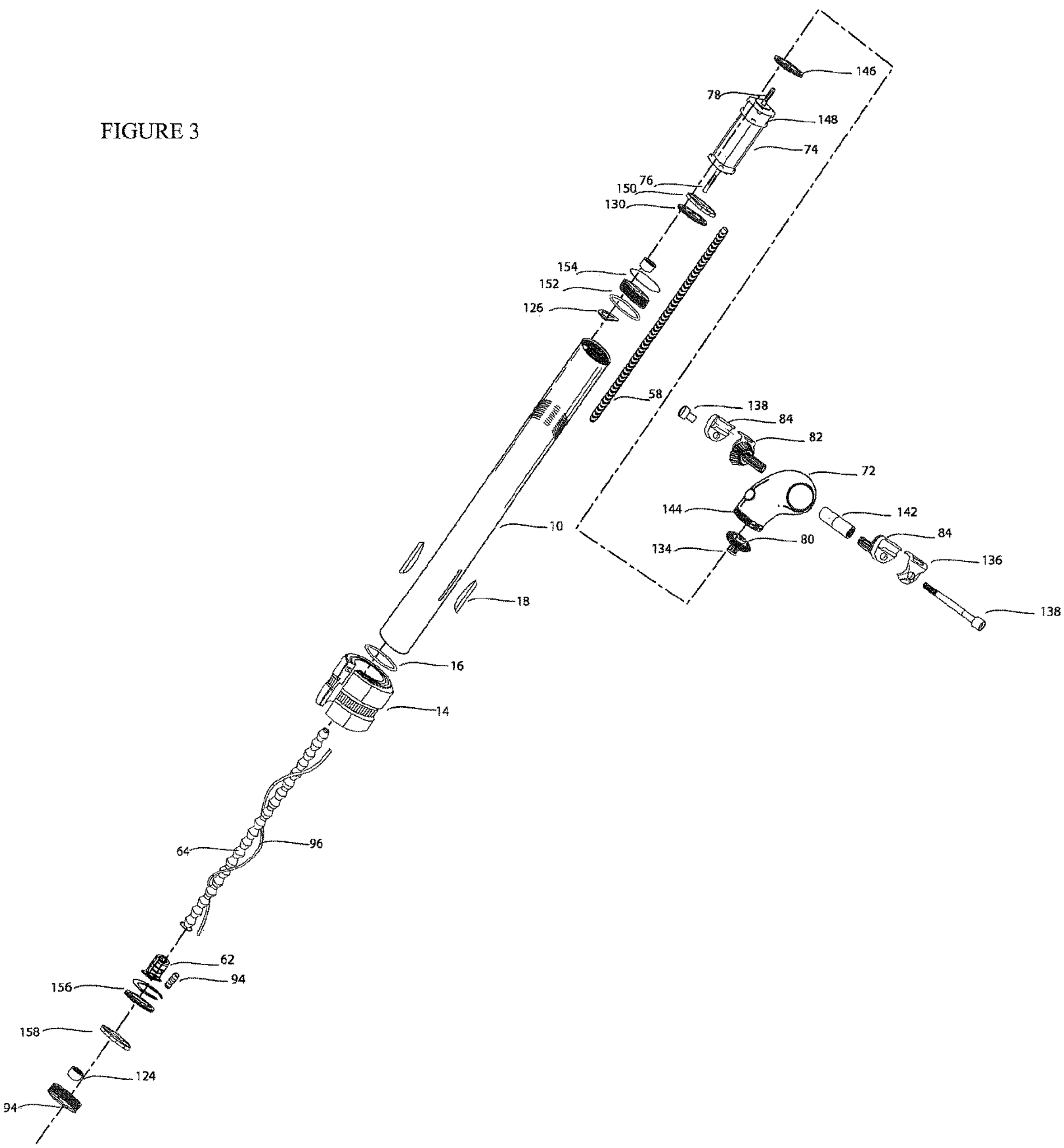
FIGURE 3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94

FIGURE 4

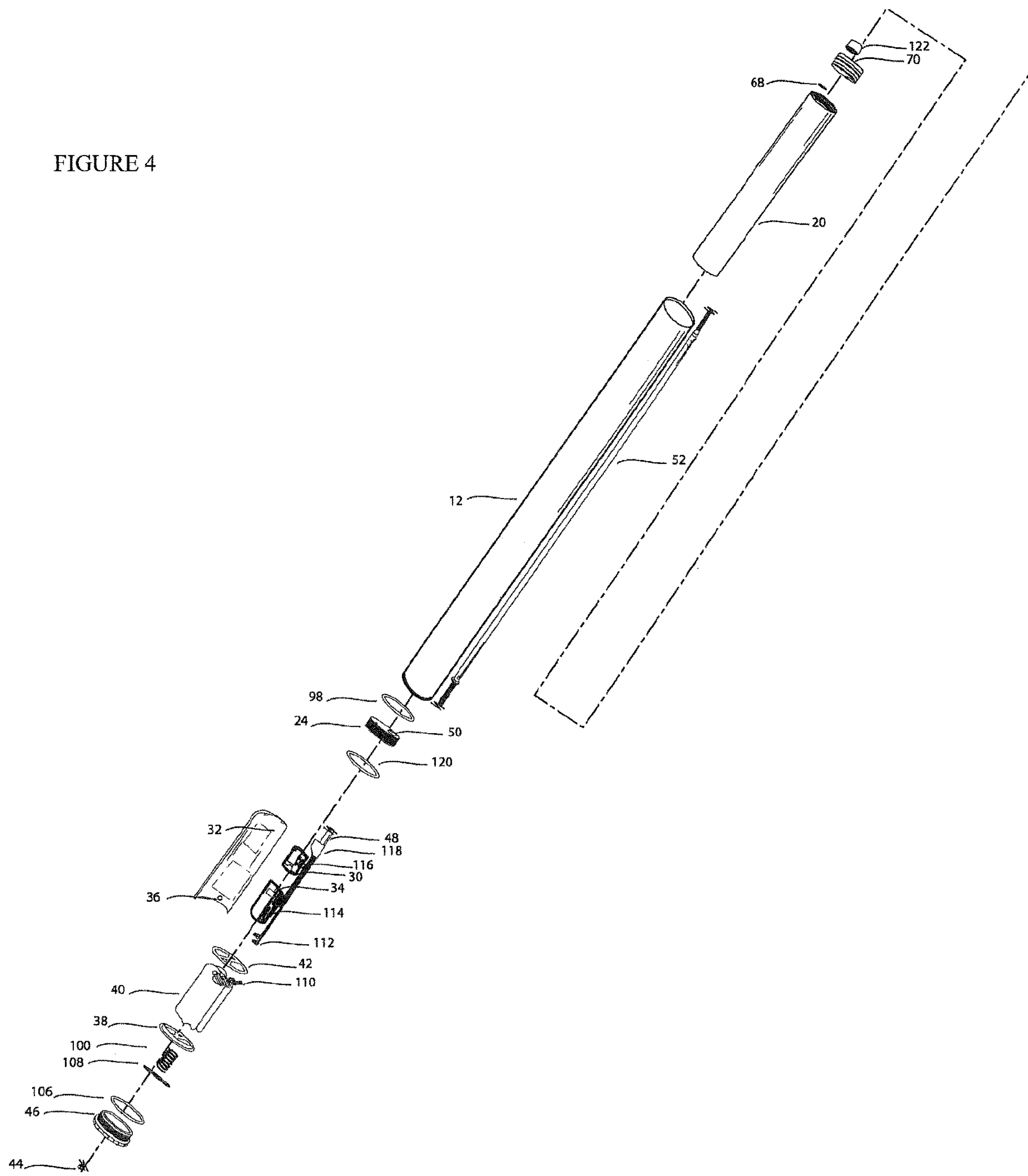
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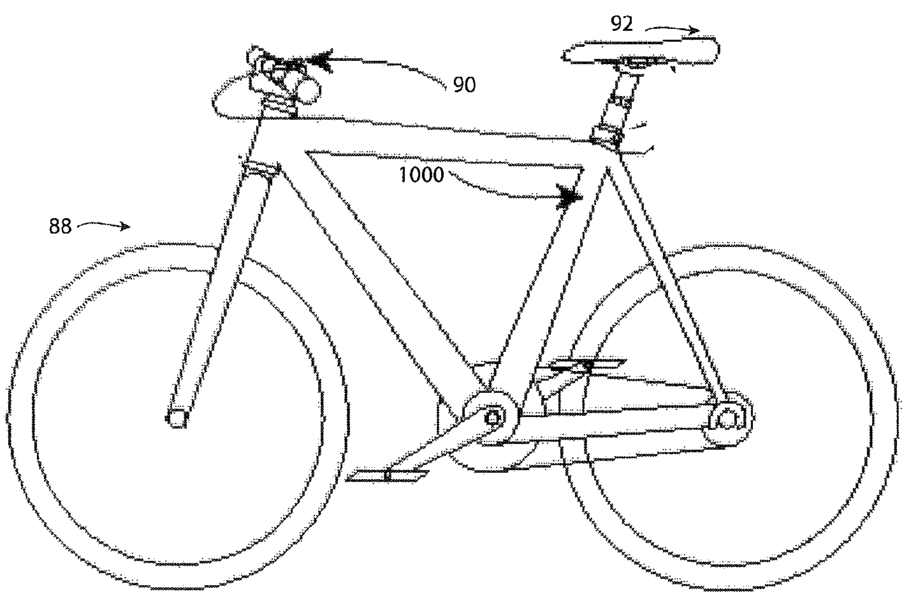
FIGURE 5



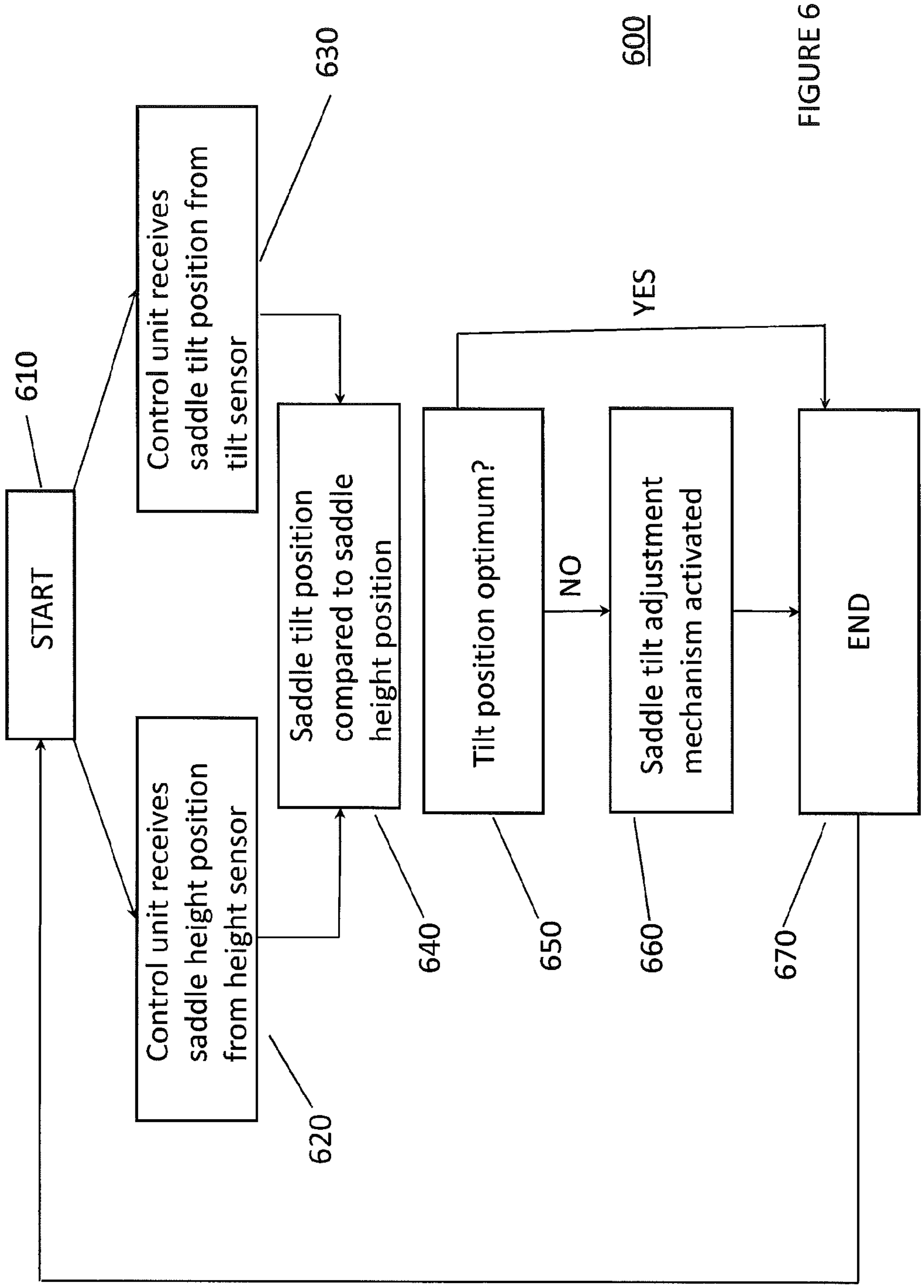
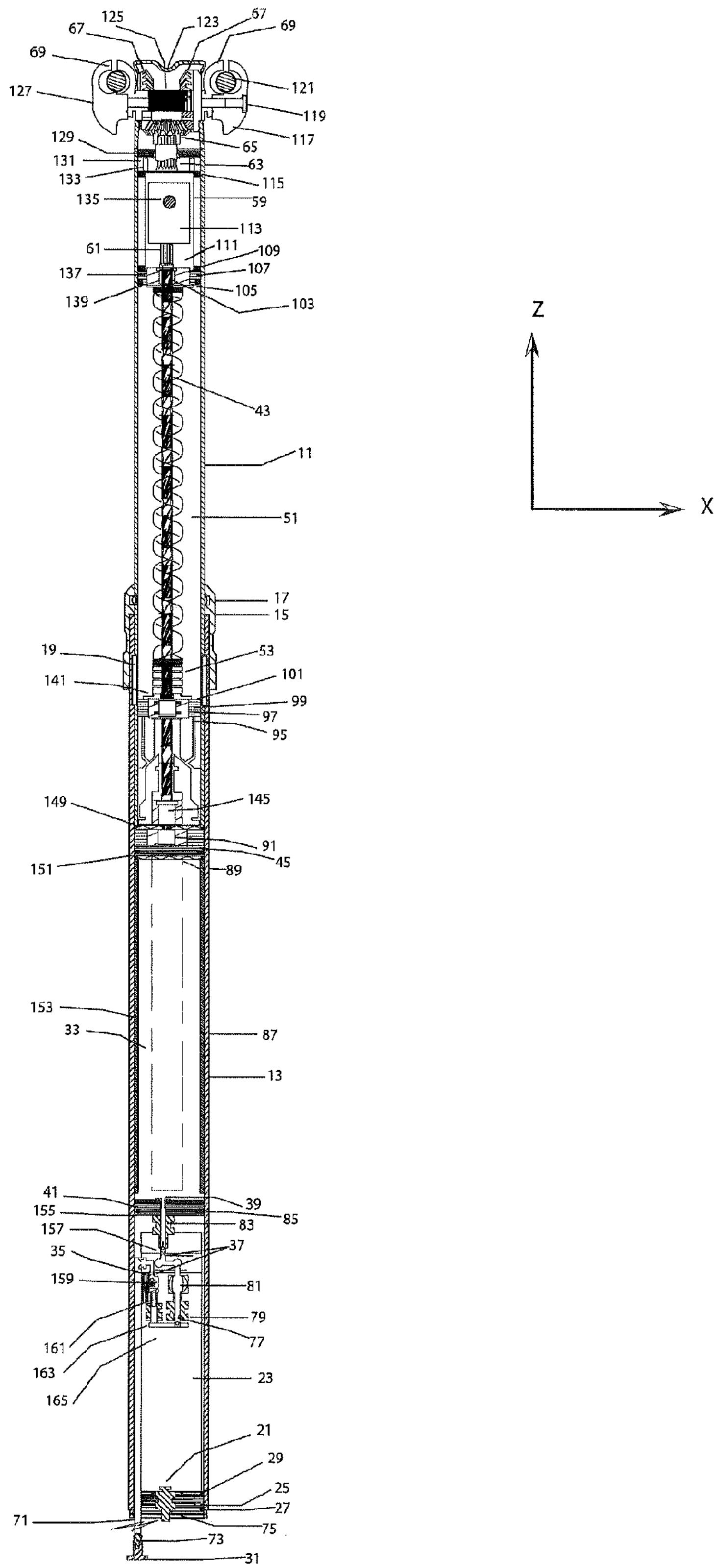

START
610
Control unit receives saddle tilt position from tilt sensor
630
Control unit receives saddle height position from height sensor
620
Saddle tilt position compared to saddle height position
640
Tilt position optimum?
650
YES
NO
Saddle tilt adjustment mechanism activated
660
END
670
600

FIGURE 6

FIGURE 7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123
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69
67
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121
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137
107
139
105
103
Z
43
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X
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141
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161
163
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73
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FIGURE 8

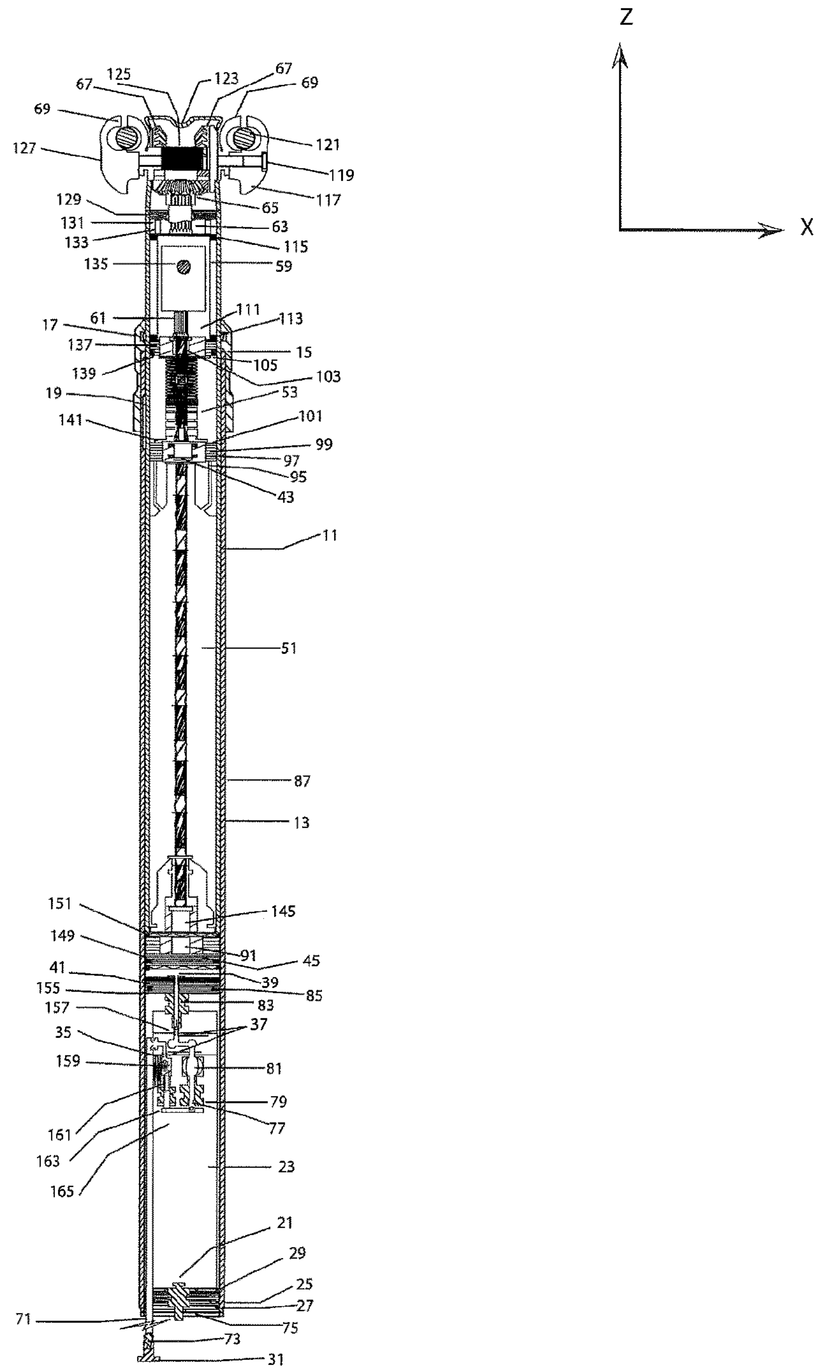

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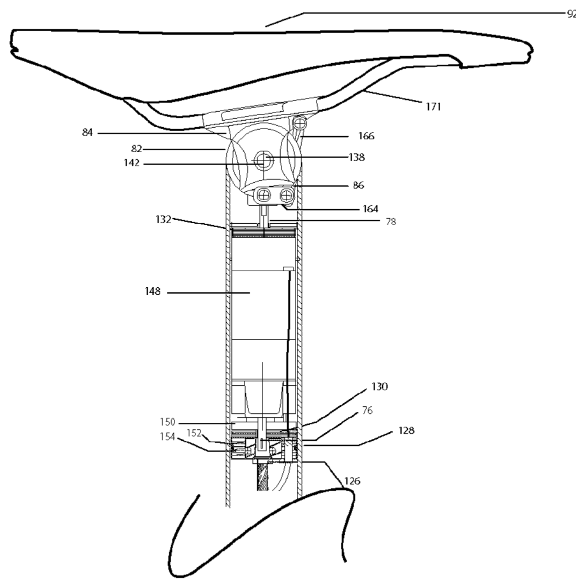
FIGURE 9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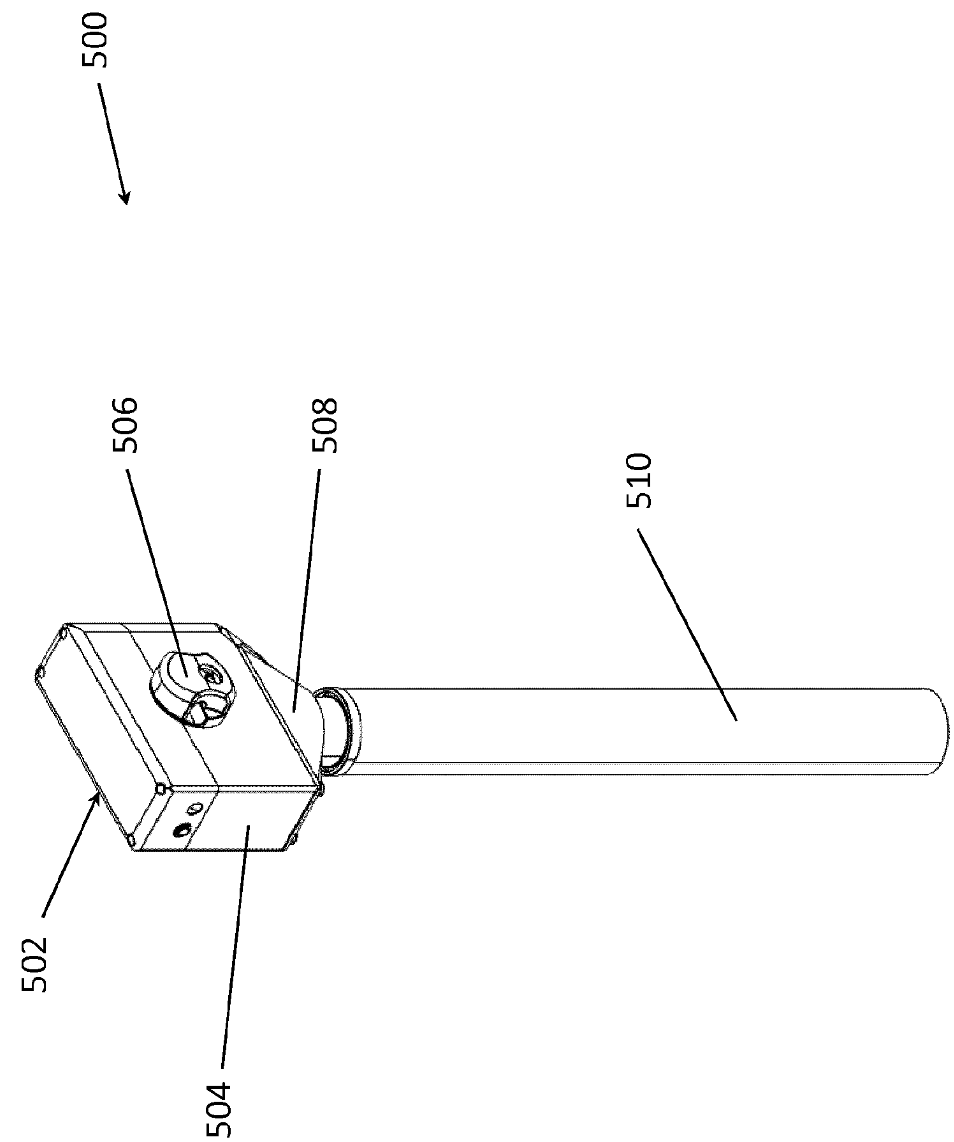


FIG. 10A

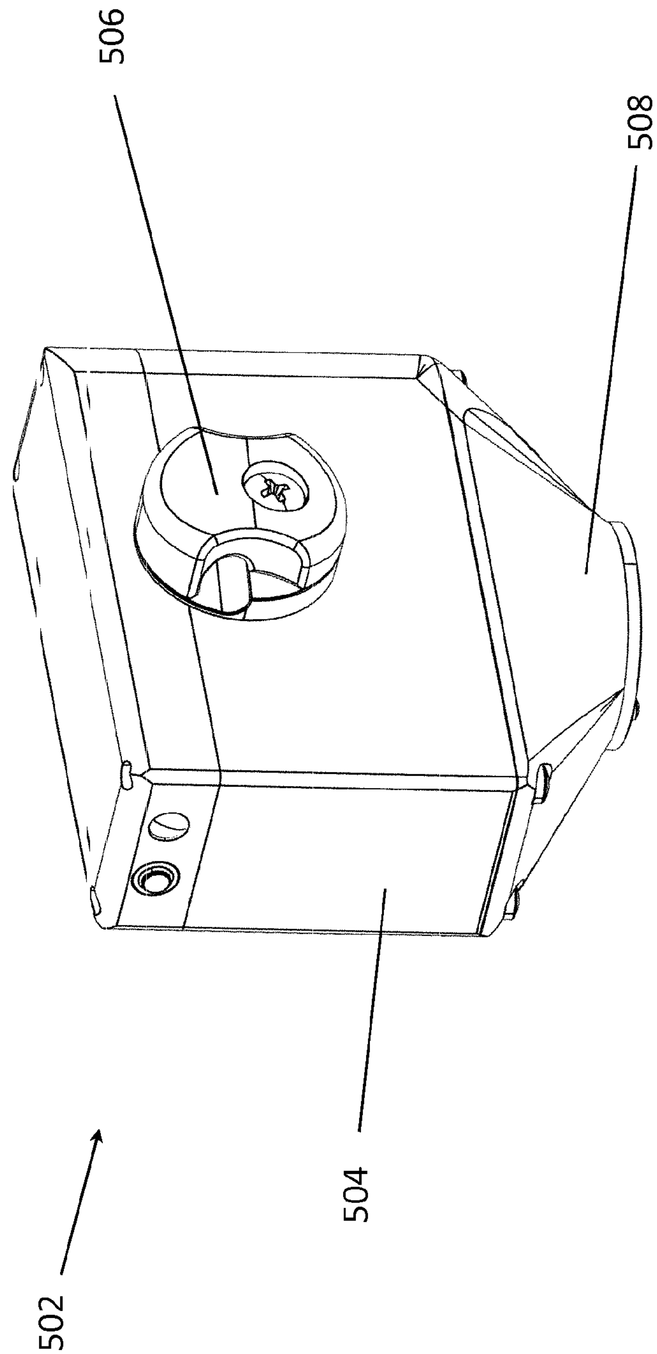


FIG. 10B

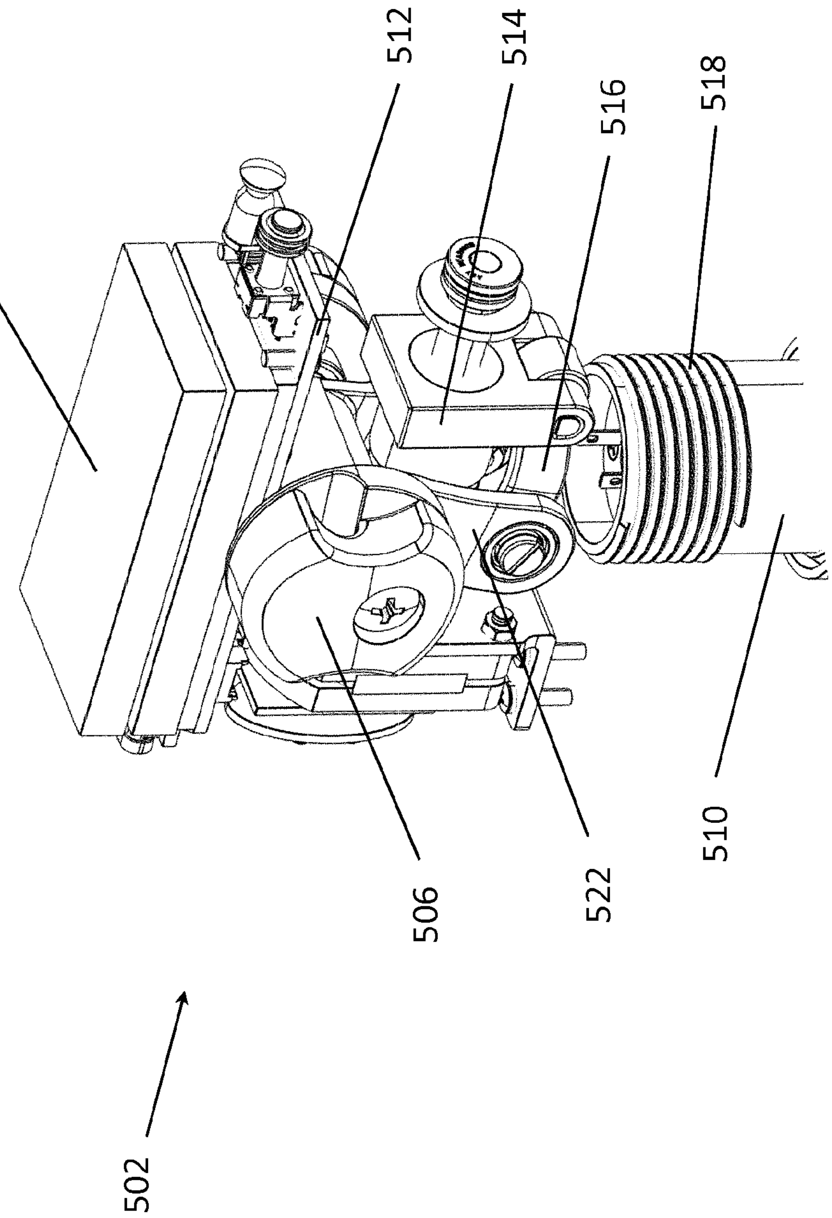


FIG. 11A

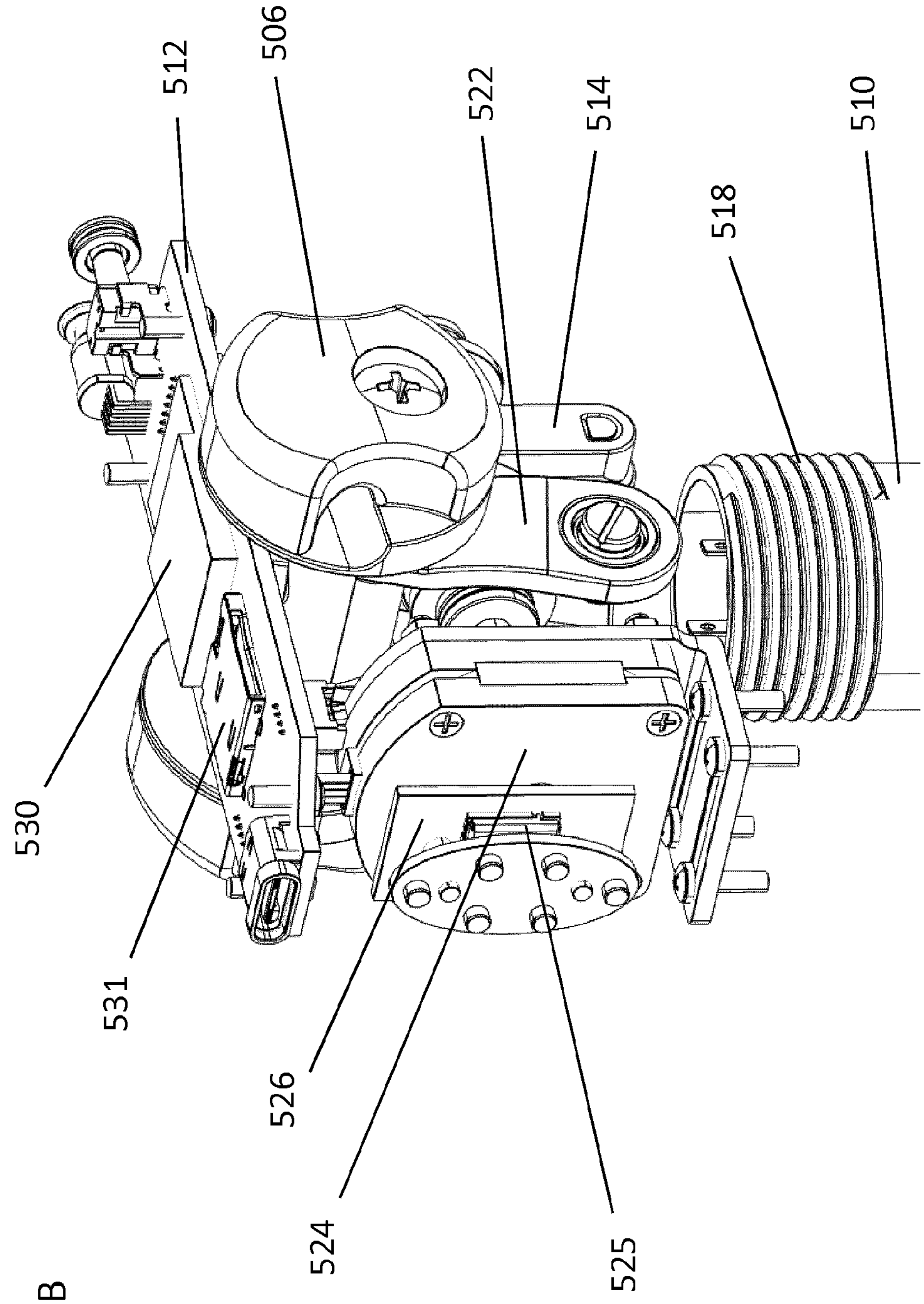


FIG. 11B

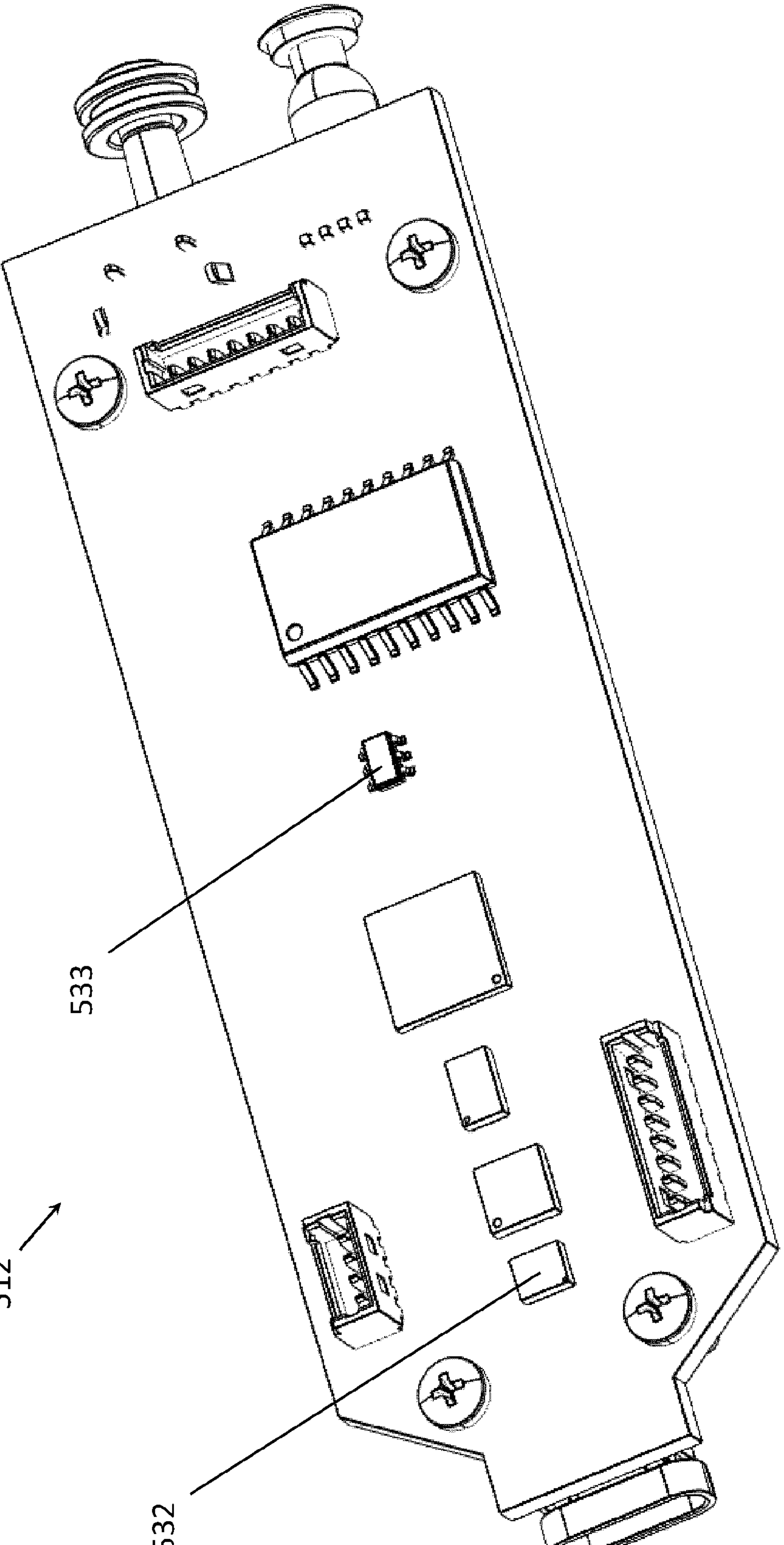


FIG. 11C

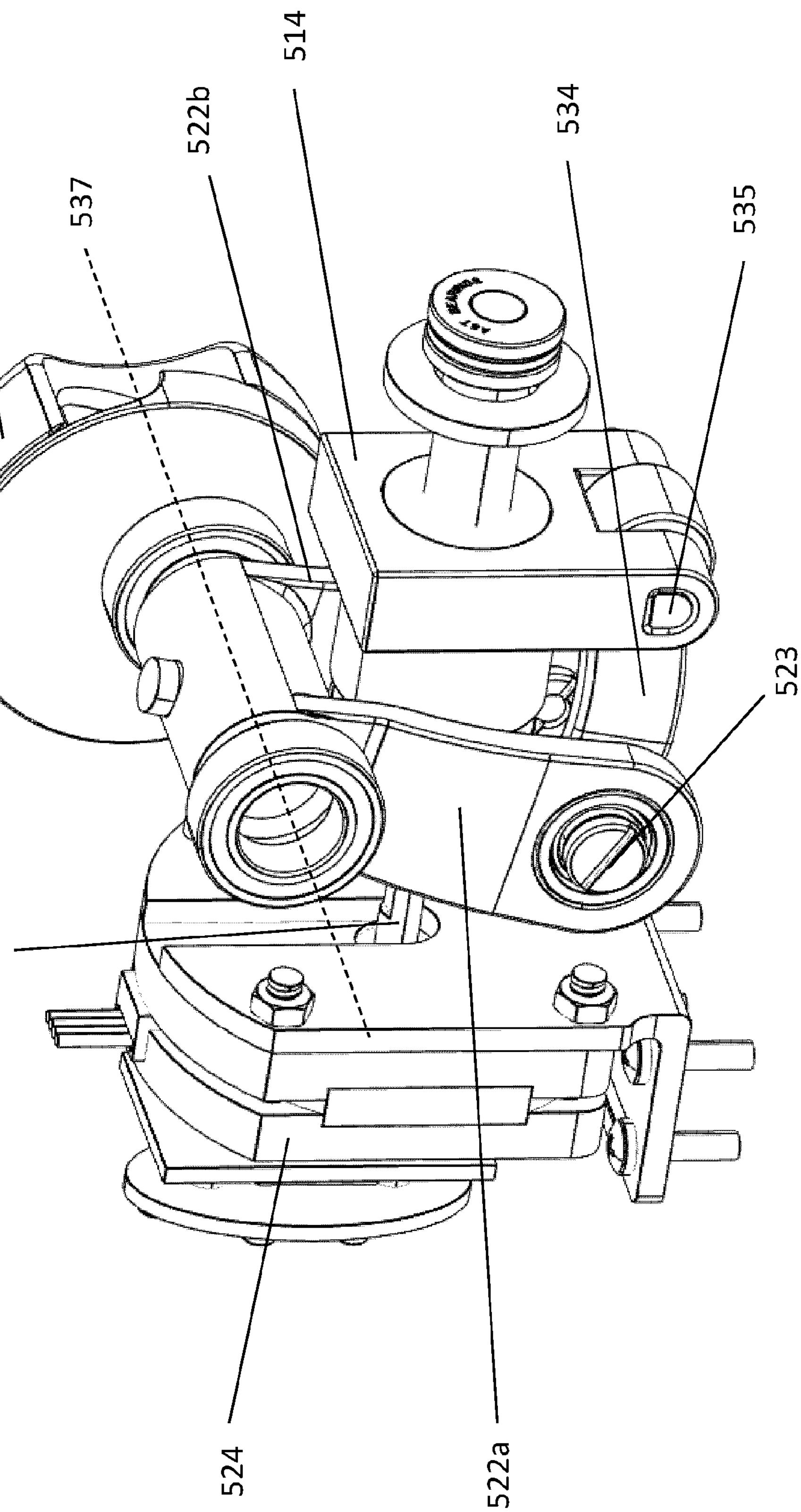


FIG. 12A

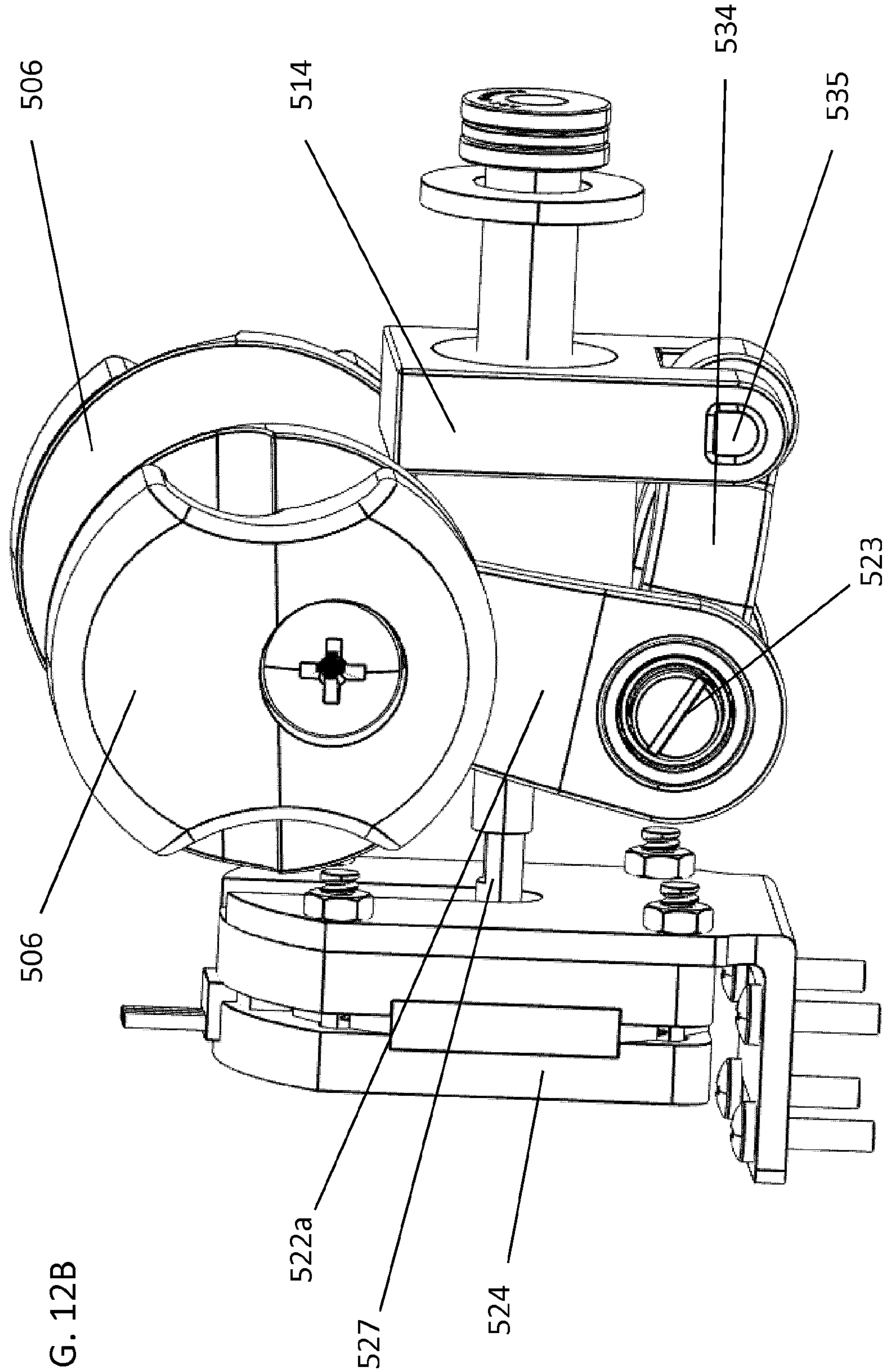


FIG. 12B

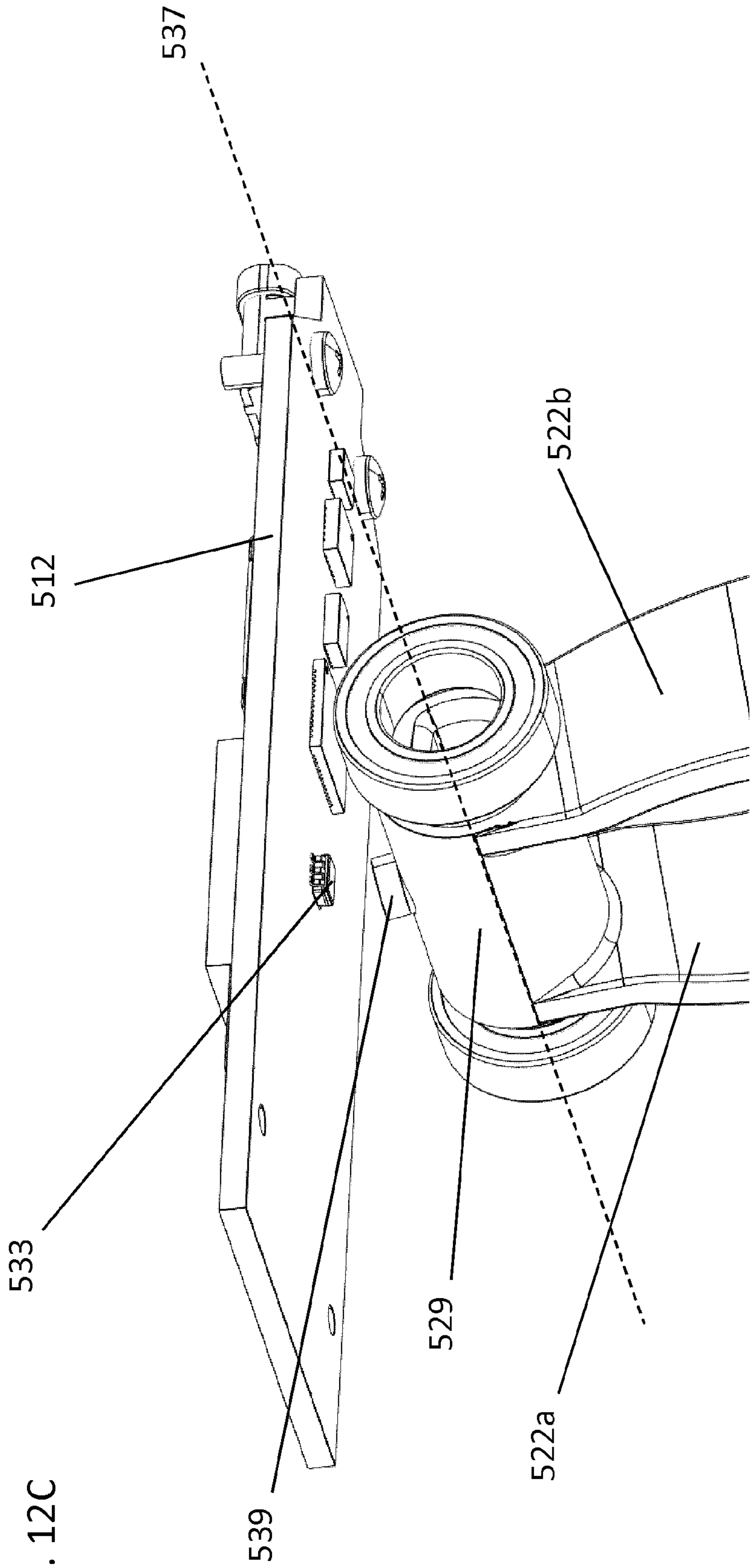
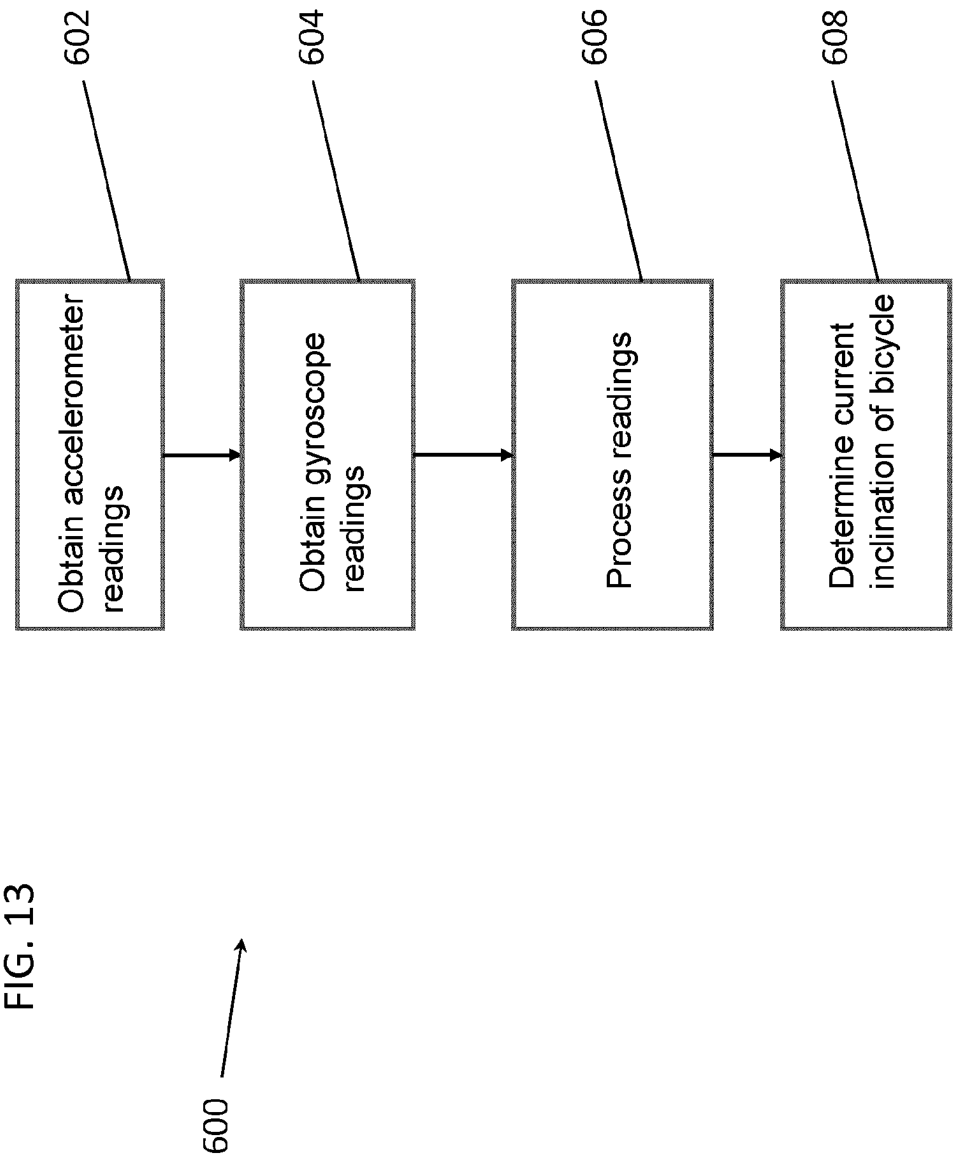


FIG. 12C



602
604
606
608
Obtain accelerometer readings
Obtain gyroscope readings
Process readings
Determine current inclination of bicycle
600

FIG. 13

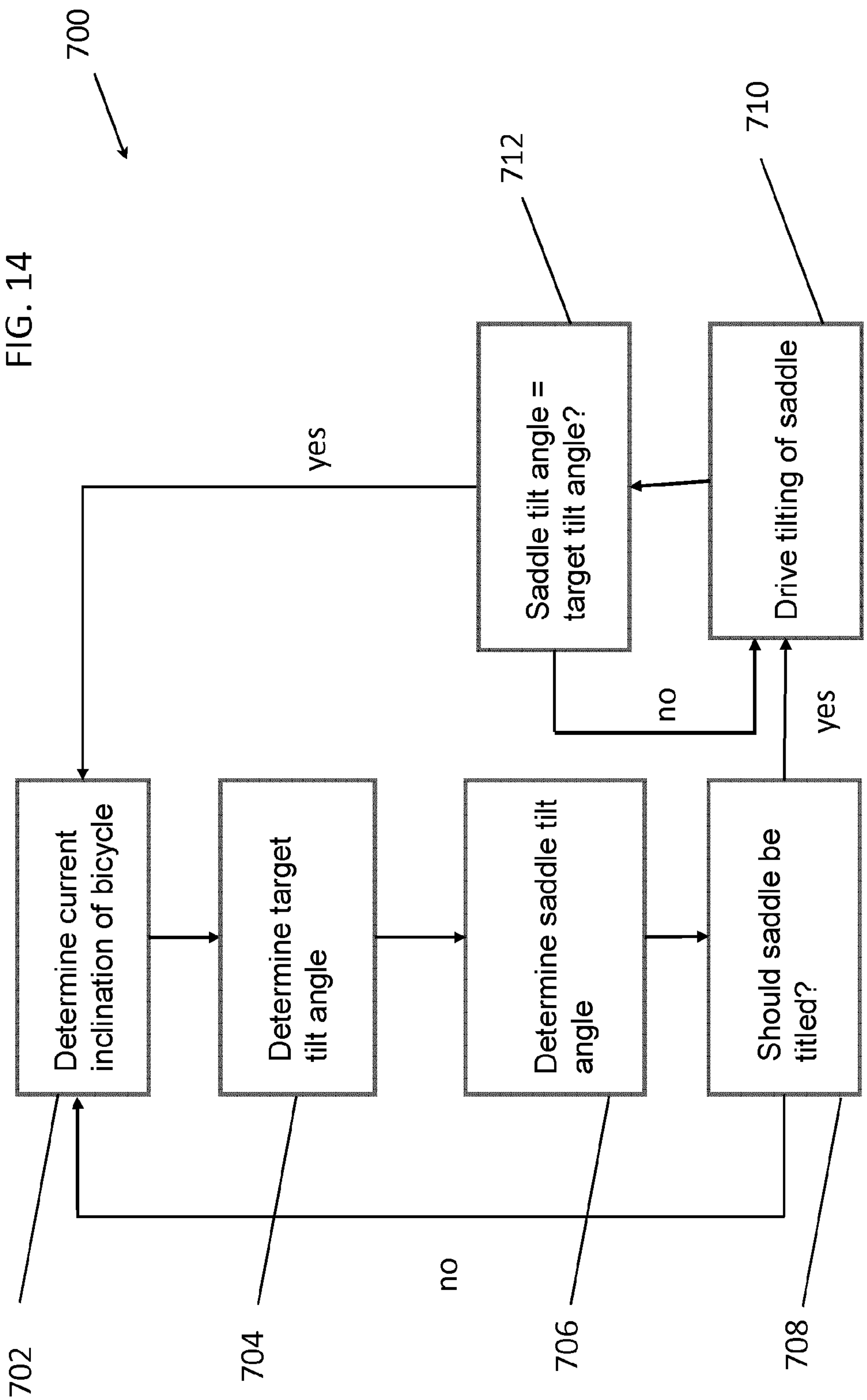

700
Determine current inclination of bicycle
Determine target tilt angle
Determine saddle tilt angle
Should saddle be titled?
Drive tilting of saddle
Saddle tilt angle = target tilt angle?
yes
no
yes
no
702
704
706
708
710
712

FIG. 14

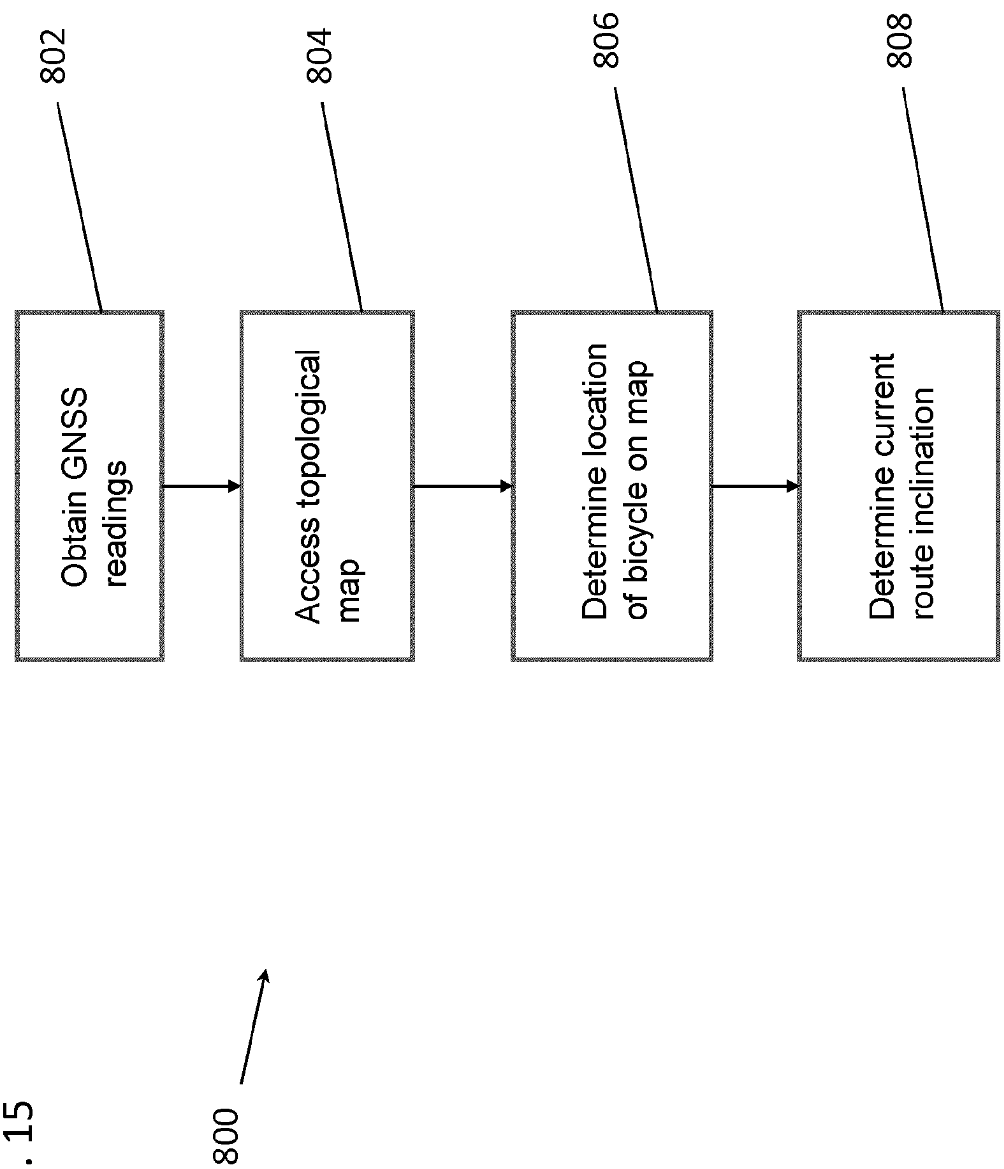

802
804
806
808
Obtain GNSS readings
Access topological map
Determine location of bicycle on map
Determine current route inclination
800

FIG. 15

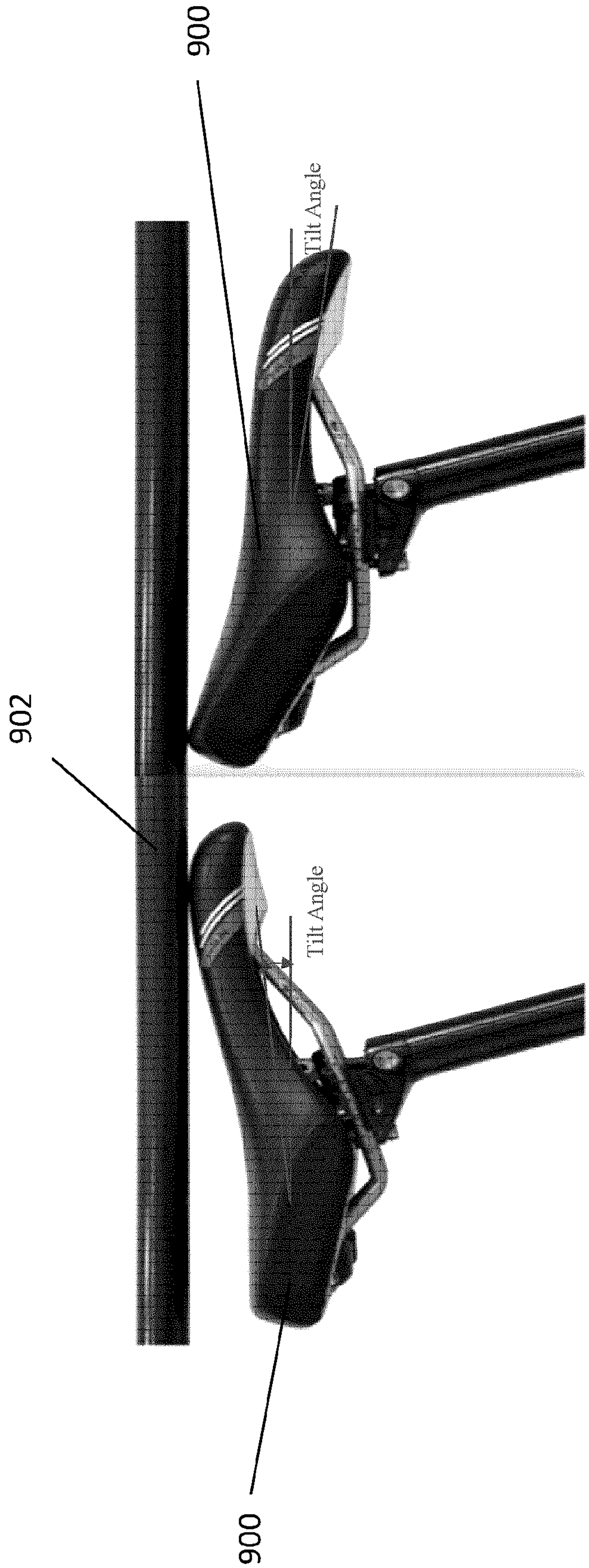


FIG. 16

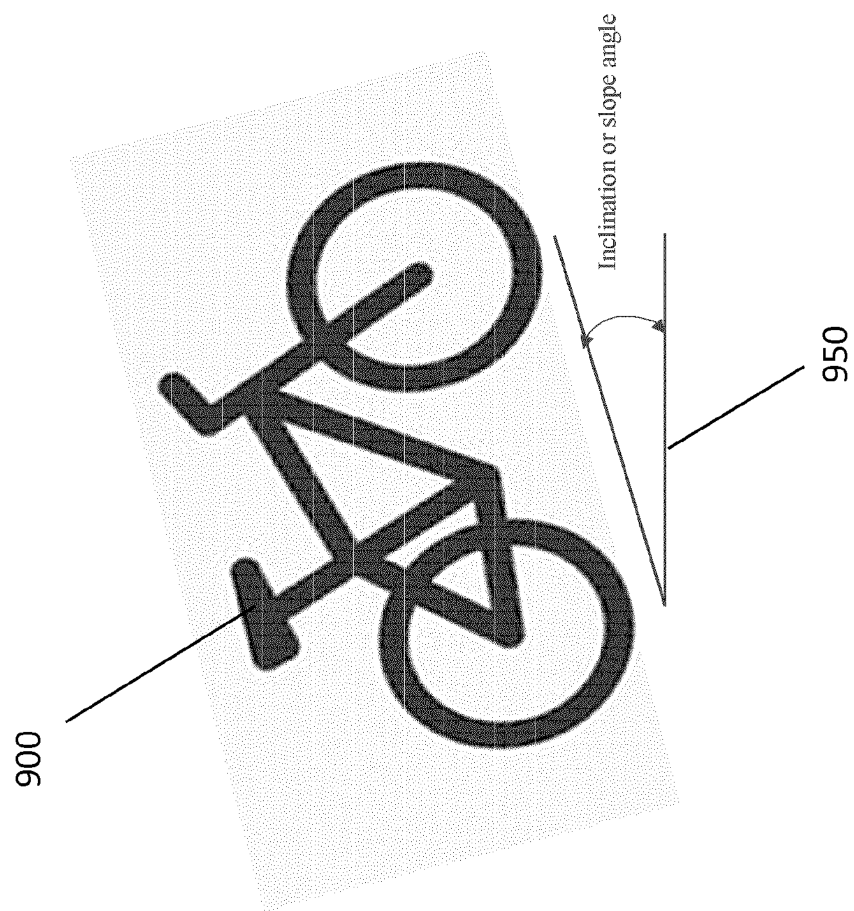


FIG. 17

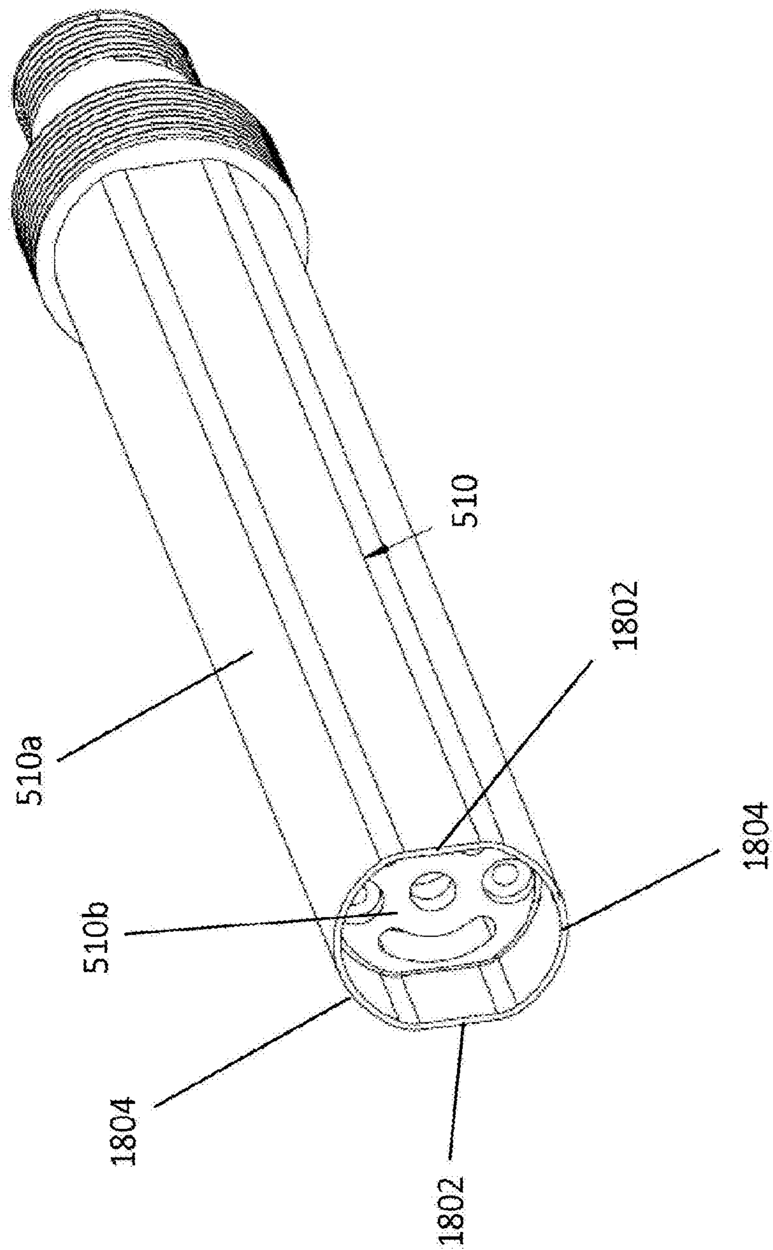


FIG. 18

FIG. 19B

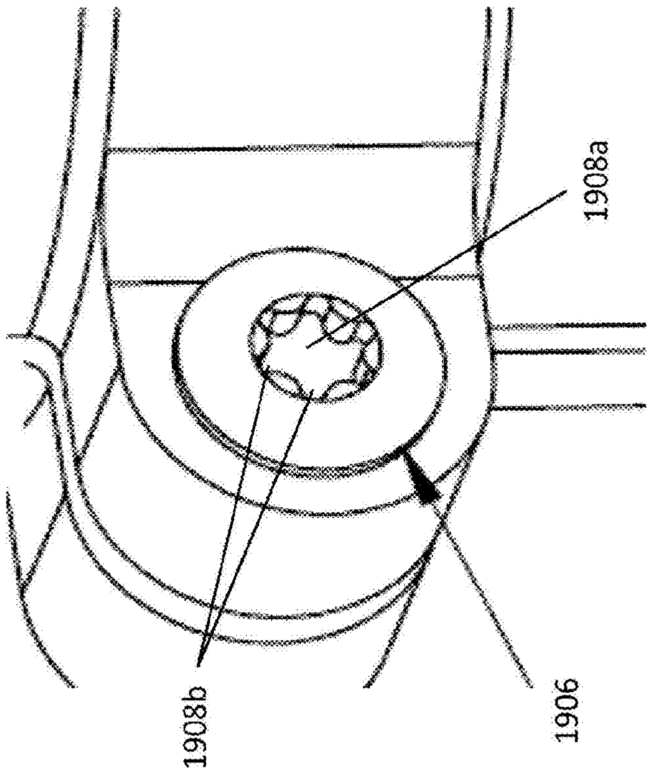
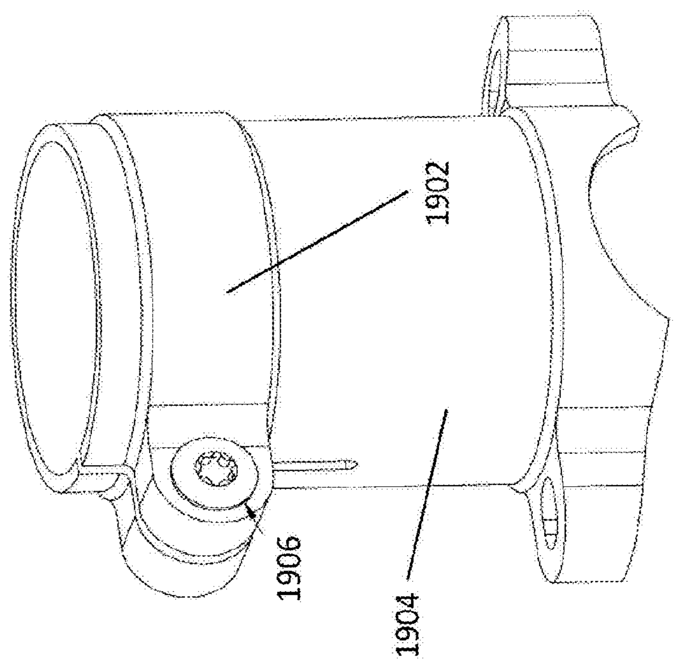


FIG. 19A



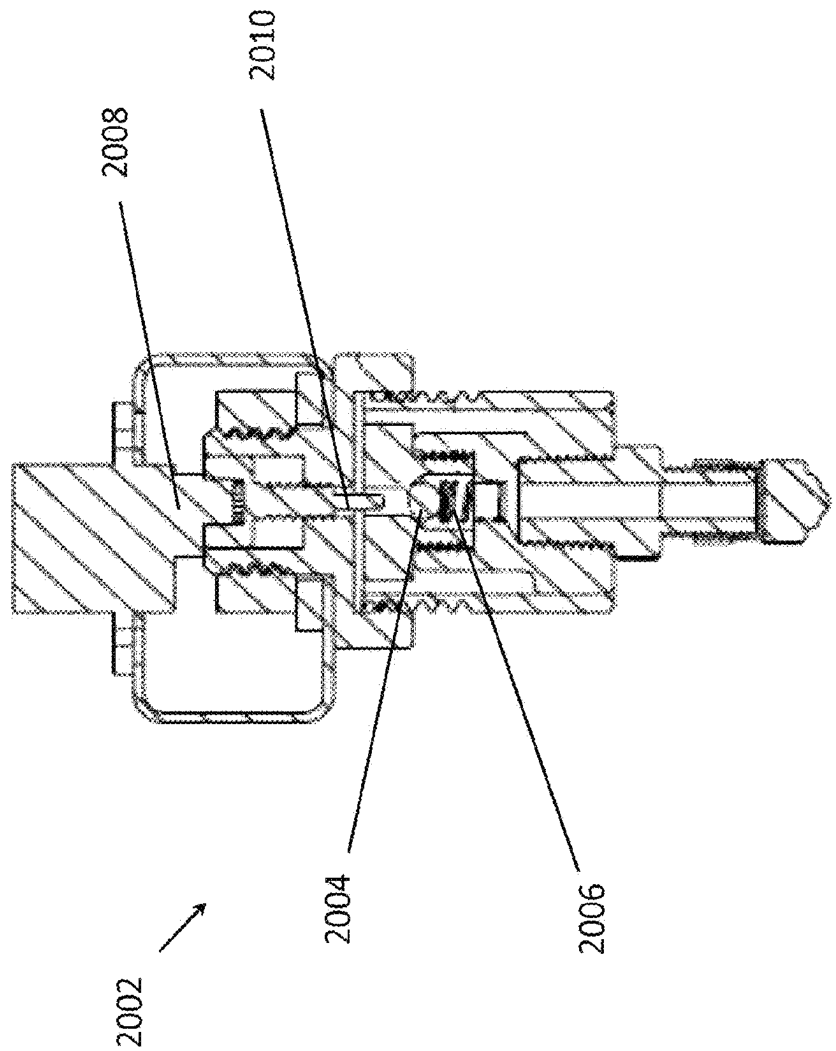


FIG. 20

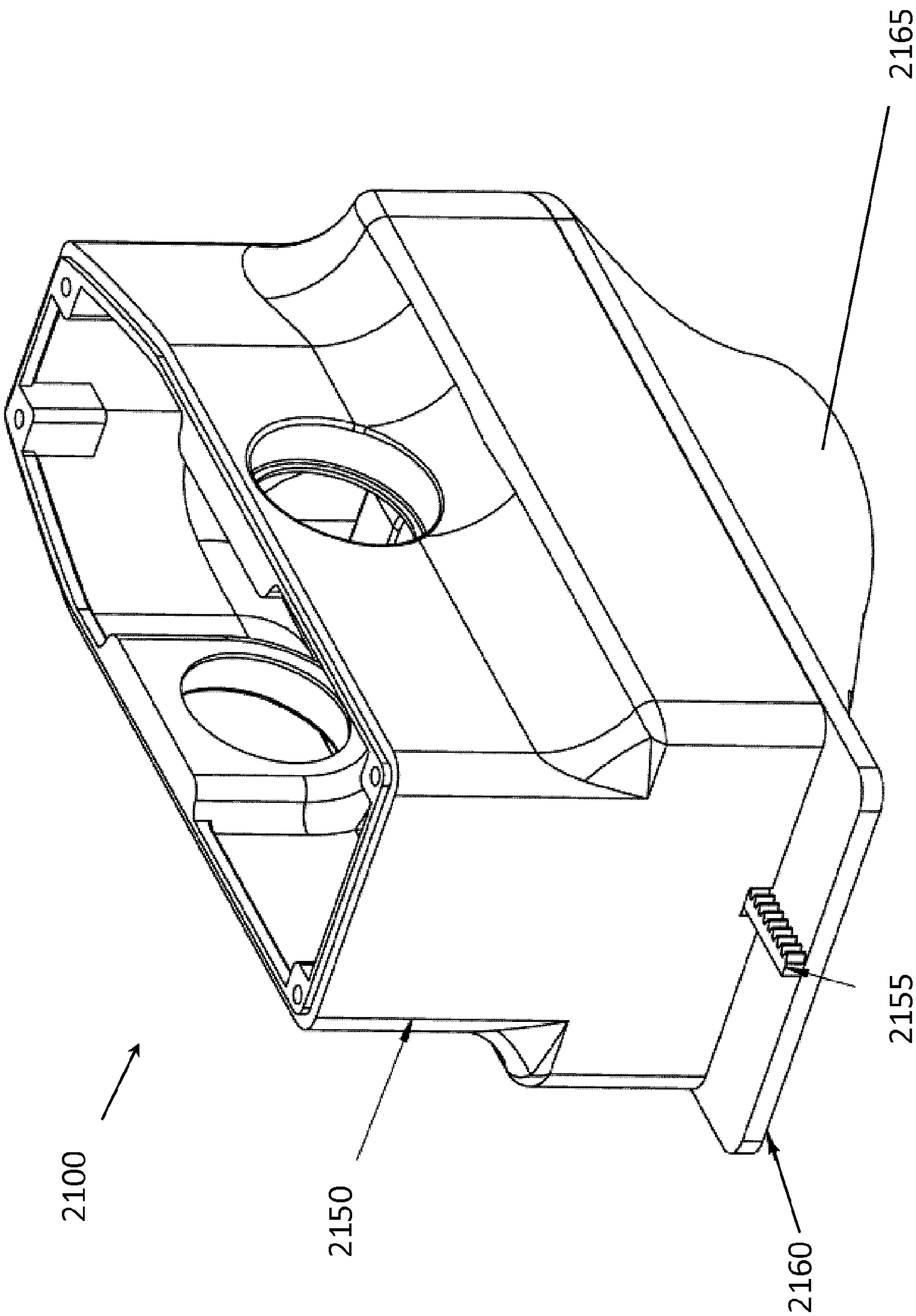


FIG. 21

DEVICE FOR ADJUSTING A SEAT POSITION OF A BICYCLE SEAT

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a U.S. National Stage entry under 35 U.S.C. § 371 of International Application No. PCT/CA2021/050623, filed on May 4, 2021, designating the United States of America and published in English on Nov. 11, 2021, which in turn claims priority to U.S. Provisional Application No. 63/019,626, filed on May 4, 2020, each of which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

The present disclosure relates to a device for adjusting a seat position, such as tilt, of a bicycle seat.

BACKGROUND TO THE DISCLOSURE

It is important for riders of bicycles to be properly positioned on their bike when cycling. Poor positioning of the rider can lead not only to decreased performance but in some cases may also lead to the development of muscular aches and pains.

One of the principal ways in which a rider can adjust their position on a bicycle is by adjusting the position of their seat (also generally referred to as a saddle). It is a known industry fact that the rider's performance is greatly enhanced if the saddle height is located closest to the optimum saddle height. The optimum saddle height is generally a function of the steepness of the surface on which the rider is cycling. Typically a rider will want a lower saddle position for descending, a mid-level saddle position for side hilling, and a raised saddle position for ascending. For example, while descending, if the saddle is in a lowered position the rider is able to move their center of gravity back over the rear wheel, to increase control and fluidity, as well as be less likely to fall forward or be thrown over the handlebars. Thus, they can descend faster and with better precision. Similarly, while ascending or climbing, if the saddle is in a raised position then the rider will now be in an extended leg position. As a result the rider will be able to get their center of gravity over parts of the rear wheel, increasing power output, control and ascending traction, as well as being less likely to fall to either side or put a foot down. Thus, they can climb faster, with less fatigue, and with improved performance and better precision.

To adjust a saddle's vertical position relative to the bicycle frame, most bicycles have a seat post that is adjustable in height. The seat post is inserted into a bicycle's seat tube to an extent that provides the desired seat height, and the seat post is subsequently clamped to the seat tube. When desiring to adjust the seat height during a ride, a rider will typically dismount, loosen the clamp, adjust the vertical position of the saddle by moving the seat post up or down, as required, and then tighten the clamp to re-engage the seat post to the seat tube.

However, manually adjusting the seat height as described above is clearly time-consuming. For example, if during a session a rider is faced with a long uphill stretch and wishes to raise the saddle, they must first dismount, make the necessary adjustment and then mount again, losing time as well as expending energy. To avoid the need to make saddle adjustments every time the steepness of the surface changes dramatically, a rider will often simply opt for a 'middle of the road' saddle height, and as a result will compromise by having the saddle too high for downhill sections, and too low for uphill sections.

Developments in the cycling industry have led to the invention of automatic saddle height adjustment mechanisms, such as those described in U.S. Pat. Nos. 6,202,971 and 7,083,180. For example, U.S. Pat. No. 6,202,971 describes seat posts that may be adjusted in height while the bicycle is in motion. Despite the introduction of 'in situ' saddle height adjustment mechanisms, there remains a need in the art to allow riders to further adjust the position of their bicycle seat while riding a bicycle.

Furthermore, while seat height is an important factor in obtaining a rider's optimal riding position, it is not the only factor. Another important parameter is seat tilt. The tilt of the seat may be considered as the angle defined by the bicycle's seat/saddle and the surface on which the bicycle is moving. In the prior art, similarly to manually adjustable seat posts, seat tilt is adjustable, when off the bike, by disengaging a locking clamp, adjusting the tilt angle of the saddle, and re-engaging the clamp. Much like manually height-adjustable seat posts, modifying the tilt of the seat in such a fashion is time-consuming and with current bicycle systems cannot be executed while the bicycle is in motion.

SUMMARY OF THE DISCLOSURE

At present a rider generally pre-sets the saddle tilt prior to riding, and during riding accepts that for certain uphill/downhill sections the seat tilt angle will not be optimal as it cannot be adjusted without dismounting the bike. Typically, the optimum tilt angle range for ascending based on rider preference is between +5° and −5°. When ascending, there is therefore a variance of approximately 10° in seat tilt between different riders. When descending, there is typically a variance of approximately 15° in seat tilt between different riders (rider preference when descending is typically between +25° and +10°). Therefore there generally exists a total variance of roughly 30° in tilt angle between riders, when descending and ascending. It would be clearly advantageous, therefore, if a device were provided that could address seat tilt without the need for a rider to dismount their bike and manually adjust the tilt of their saddle.

The present disclosure provides a device that may allow for automatic adjustment of the seat tilt of a bicycle during riding of the bicycle (i.e. when the bicycle is in motion). The device may furthermore allow for automatic adjustment of seat height, substantially at the same time that seat tilt is being adjusted. Riders may automatically position themselves in the anatomically correct expert position via a seat post that may be automatically adjustable in height in combination with a seat whose tilt may be automatically adjusted, while the bicycle is in motion. The controller for actuating the seat height and seat tilt adjustments may be within easy reach of the rider (for example the controller may be located on a handlebar of the bike).

In accordance with an aspect of the disclosure, there is provide a device for adjusting a seat position of a bicycle seat. The device comprises a seat tube coupling configured to couple to a bicycle seat tube; a seat coupling configured to couple to a bicycle seat; and a seat adjustment mechanism movably coupling the seat tube coupling and the seat coupling. The seat adjustment mechanism comprises a tilt actuator operable to adjust a tilt of the seat coupling relative to the seat tube coupling. The device further comprises a tilt controller remote from and communicative with the seat adjustment mechanism and operable by a rider of a bicycle to actuate the tilt actuator thereby adjusting the tilt of the seat coupling relative to the seat tube coupling.

Thus, a rider may adjust their seat tilt while riding the bike, without the need to dismount. Riders may therefore be able to achieve better control, style, and riding technique, while also increasing safety and enjoyment, without expending significant time and energy dismounting and manually adjusting seat tilt, as in the prior art. Riders may furthermore no longer have to compromise with seat tilt set to 'middle of the road' settings and may enjoy the benefit of an optimum seat tilt simply by actuating the seat adjustment mechanism.

The seat adjustment mechanism may be arranged to provide tilting of the seat coupling relative to the seat tube coupling. The seat coupling may be any mechanical coupling configured to couple to or engage with a bicycle seat. The seat tube coupling may be any mechanical coupling configured to couple to or engage with a seat tube of a bicycle. For example, the seat tube coupling may comprise a seat post arranged to be received within a bicycle seat tube. The tilt actuator may be arranged when actuated to cause tilting of the seat coupling along an axis, thereby causing corresponding tilting of a bicycle seat attached to the seat coupling. The tilt controller may be electrically, mechanically, or hydraulically communicative with the seat adjustment mechanism. Other forms of communication are envisaged between the tilt controller and the seat adjustment mechanism. The tilt controller is actuable by a rider when the bicycle is in motion. For example, the tilt controller may take the form of a lever, button or the like, and may be positioned on the handlebars of the bicycle, and the rider may use their thumb or another digit to activate the tilt controller. The tilt controller may be located at other points on the bicycle, provided that they may be relatively easily accessed by the rider during riding of the bicycle.

The tilt actuator may comprise a seat coupling gear fixed to the seat coupling. The seat adjustment mechanism may further comprise a prime mover rotatably coupled to the seat coupling gear along a tilt axis such that movement of the prime mover rotates the seat coupling gear thereby adjusting the tilt of the seat coupling along the tilt axis.

The prime mover may comprise an electrical motor or a source of pressurized air. Actuation of the tilt controller may cause operation of the electrical motor, or may cause release of the pressurized air.

The prime mover may comprise a rotatable upper shaft comprising a drive gear. The tilt axis and upper shaft may be perpendicular to each other, and the drive gear and seat coupling gear may be bevelled and coupled to each other such that rotation of the upper shaft causes rotation of the seat coupling gear.

The seat adjustment mechanism may further comprise a linear actuator operable to linearly translate the seat tube coupling relative to the seat coupling. The seat adjustment mechanism may further comprise a height controller remote from and communicative with the seat adjustment mechanism and operable by the rider to actuate the linear actuator thereby adjusting the height of the seat coupling relative to the seat tube coupling.

The rider may therefore, using the height controller, actuate the linear actuator to adjust a height of the bicycle seat 'on the fly' (i.e. while the bicycle is in motion). The linear actuator may comprise a shaft, such as a ball screw or nut, threadedly engaged with a seat post or other linear axial assembly. For example the ball screw may be threadedly engaged with a ball nut fixed to the seat post. Operation of the height controller may provide height adjustment of the seat coupling relative to the seat tube coupling. The height adjustment may comprise telescoping or translating of one or more seat posts relative to one or more further seat posts.

The height controller may be electrically, mechanically, or hydraulically communicative with the seat adjustment mechanism. The height controller may be actuable by a rider when the bicycle is in motion. For example, the height controller may take the form of a lever, button or the like, and may be positioned on the handlebars of the bicycle, and the rider may use their thumb or another digit to activate the height controller.

The tilt controller and/or height controller could be buttons, paddles, levers, or similar, and may be designed with different heights from the assembly that houses them. A rider may therefore not need to look at the controllers when riding, as the controllers may be operated by touch rather than necessarily by sight. In some embodiments, the rider may operate the controllers through other means, such as by voice command, or via a heads-up display. In still other embodiments, the controllers may be configured to activate according to autonomous pre-sets.

The seat adjustment mechanism may further comprise a lower post coupled to the seat tube coupling and defining a translation axis; and an upper post coupled to the seat coupling and translatable relative to the lower post along the translation axis. The prime mover may be coupled to the linear actuator. The linear actuator may be coupled to the lower post and upper post and actuable by the prime mover to linearly translate the lower post relative to the upper post.

Thus, the tilt actuator may translate relative to the lower post during telescoping/translating of the upper post relative to the lower post.

The prime mover may comprise at least one electrical motor having a first drive shaft rotatably coupled to the seat coupling gear. The tilt controller may be communicative with the at least one electrical motor.

The rider may therefore, using the tilt controller, actuate the electrical motor to operate the tilt actuator, thereby adjusting a tilt of the bicycle seat 'on the fly' (i.e. while the bicycle is in motion), via tilting of the seat coupling relative to the seat tube coupling. The electrical motor may be arranged to simultaneously operate both tilting and translation of the seat coupling relative to the seat tube coupling. Alternatively, the motor may be arranged to operate tilting of the seat coupling relative to the seat tube coupling independently of translation of the seat coupling relative to the seat tube coupling.

The linear actuator may comprise a threaded lower shaft in rotatable threaded engagement with one of the upper post and the lower post. The at least one electrical motor may be fixed to the other one of the upper post and lower post. The at least one electrical motor may comprise a second drive shaft coupled to the threaded lower shaft and operable to rotate the threaded lower shaft thereby causing the upper post to translate relative to the lower post.

Through the threaded engagement of the threaded lower shaft with one of the upper post and the lower post, rotation of the threaded lower shaft may be arranged to cause the post to which the lower shaft is threadedly coupled to translate relative to the threaded lower shaft. The other of the upper and lower posts may translate with the threaded lower shaft as the threaded lower shaft is rotated.

The linear actuator may further comprise a threaded ball nut fixed to the one of the lower post and the upper post. The threaded lower shaft may be a threaded ball screw rotatably engaging the threaded ball nut.

More generally, the linear actuator may comprise any other suitable linear motion assembly configured to convert motion of the second drive shaft of the at least one electrical motor into corresponding linear motion of the upper post relative to the lower post. For example, the linear actuator may comprise any other suitable linear actuator such as an offset linear actuator or a direct drive linear actuator. Such linear actuators may include tooth belts, rails, rods, or block and tackle configurations. Furthermore, in the above-described embodiments, instead of a threaded ball nut, a roller nut may be used, and, instead of a threaded ball screw, a roller screw may be used. An arbor may also be used. The arbor may be used with a sleeve in a chuck-like assembly to couple to the linear actuator. The arbor may facilitate modification of the thread grade, tooth grade, etc., of the linear actuator.

The tilt controller may comprise a tilt control interface mountable to a portion, such as a handlebar, of the bicycle, such that the tilt control interface may be operable by the rider to actuate the tilt actuator when the bicycle is in motion. In some embodiments, the tilt control interface may be mountable to the rider.

The height controller may comprise a height control interface mountable to a portion, such as a handlebar, of the bicycle, such that the height control interface may be operable by the rider to actuate the linear actuator when the bicycle is in motion. In some embodiments, the height control interface may be mountable to the rider.

One controller (tilt or height) may be operated independently of the other (height or tilt, respectively). In the one embodiment, this may be achieved for example by providing two motors: one to control height and the other to control tilt. Alternatively, one motor with two independently controllable outputs may be used.

The tilt controller and the height controller may be integrated and may comprise a combined tilt and height control interface mountable to a portion, such as a handlebar, of the bicycle and which is operable by the rider to substantially simultaneously actuate the tilt actuator and the linear actuator when the bicycle is in motion. In some embodiments, the combined tilt and height control interface may be mountable to the rider. Thus, actuation of the tilt and height controllers (integrated in the combined tilt and height control interface) may cause substantial simultaneous actuation of both the tilt actuator and the linear actuator. This may result in both adjustment of a bicycle seat tilt and a bicycle seat height. In particular, raising of the saddle height may result in corresponding downward tilting of the saddle. Similarly, lowering of the saddle height may result in corresponding upward tilting of the saddle. The ratio of height adjustment to tilt adjustment may be pre-set during manufacture of the device, and may be adjustable by a user.

The prime mover may comprise at least one electrical motor which drives the first and second drive shafts. The combined tilt and height control interface may be communicative with the at least one electrical motor.

The prime mover may comprise a pressurized air chamber mounted in the lower post. The linear actuator may comprise a piston assembly with a piston chamber in the lower post fluidly coupled to the pressurized air chamber via an air valve. The linear actuator may further comprise a piston fixed to the upper post and movable within the lower post along the translation axis. The height controller may be further communicative with the air valve and operable to open the air valve to enable air to pass between the pressurized air chamber and the piston thereby linearly translating the upper post relative to the lower post.

The rotatable upper shaft may be fixed to the upper post along the translation axis. The rotatable upper shaft may be in rotatable threaded engagement with the lower post, such that linear translation of the upper post relative to the lower post may cause the rotatable upper shaft to rotate.

The tilt actuator may further comprise a gear reduction unit rotatably coupling the rotatable upper shaft to the seat coupling gear. The gear reduction unit may be coupled to the rotatable upper shaft, such that rotation of the rotatable upper shaft may provide an input to the gear reduction unit which, as an output, provides a reduced rotation of the seat coupling gear.

The device may further comprise a resilient bias (such as a compression spring) arranged to bias the upper post away from the lower post along the translation axis. The resilient bias may assist the upper post in translating relative to the lower post, thereby making it easier for the rider to raise the saddle when using a source of pressurized air (as opposed to a motor) to adjust the height of the bicycle seat. Additionally, the resilient bias may provide a dampening force to control the descent of the upper post relative to the lower post when the rider wishes to lower the saddle.

The device may further comprise a height sensor arranged to determine a height of the seat coupling relative to the seat tube coupling. The device may further comprise a tilt sensor arranged to determine a tilt of the seat coupling relative to the seat tube coupling. The device may further comprise a control unit communicative with the height sensor and the tilt sensor. The control unit may be arranged, based on one of the determined height and tilt of the seat coupling relative to the seat tube coupling, to actuate one of tilt actuator and the linear actuator, respectively.

The height sensor may be any sensor configured to detect a change in height of the seat coupling relative to the seat tube coupling. For example, the height sensor may be positioned and arranged to detect a degree of translation or telescoping of one tube or post of the device relative to another tube or post of the device. The tilt sensor may be any sensor configured to detect a change in the angle of tilt of the seat coupling relative to the seat tube coupling. For example, the tilt sensor may be positioned and arranged to detect a degree of rotation of a gear relative to a reference point.

According to a further aspect of the disclosure, there is provided a device for adjusting a seat position of a bicycle seat. The device comprises a seat tube coupling configured to couple to a bicycle seat tube; a seat coupling configured to couple to a bicycle seat; and a seat adjustment mechanism comprising a tilt actuator operable to adjust, about a tilt axis, a tilt of the seat coupling relative to the seat tube coupling. The tilt actuator comprises a seat coupling gear fixed to the seat coupling. The seat adjustment mechanism further comprises a prime mover movably coupled to the seat coupling gear such that movement of the prime mover causes rotation of the seat coupling gear thereby adjusting the tilt of the seat coupling about the tilt axis. The device further comprises a tilt controller remote from and communicative with the seat adjustment mechanism and operable by a rider of a bicycle to actuate the tilt actuator thereby adjusting the tilt of the seat coupling relative to the seat tube coupling.

The prime mover may comprise a rotatable upper shaft comprising a drive gear. The tilt axis and upper shaft may be perpendicular to each other. The drive gear and seat coupling gear may be coupled to each other such that rotation of the upper shaft causes rotation of the seat coupling gear.

The drive gear and seat coupling gear may be bevelled, toothed, or may incorporate one or more bearings, such as ½ moon, sealed, or needle bearings.

The seat adjustment mechanism may further comprise a linear actuator operable to linearly translate the seat tube coupling relative to the seat coupling. The device may further comprise a height controller remote from and communicative with the seat adjustment mechanism and operable by the rider to actuate the linear actuator thereby adjusting the height of the seat coupling relative to the seat tube coupling.

The seat adjustment mechanism may further comprise: a lower post coupled to the seat tube coupling and defining a translation axis; and an upper post coupled to the seat coupling and translatable relative to the lower post along the translation axis. The prime mover may be coupled to the linear actuator, the linear actuator may be coupled to the lower post and upper post, and the linear actuator may be actuable by the prime mover to linearly translate the lower post relative to the upper post.

The prime mover may comprise at least one motor having a first drive shaft movably coupled to the seat coupling gear. The tilt controller may be communicative with the at least one motor.

The seat adjustment mechanism may further comprise a linear actuator operable to linearly translate the seat tube coupling relative to the seat coupling. The device may further comprise a height controller remote from and communicative with the seat adjustment mechanism and operable by the rider to actuate the linear actuator thereby adjusting the height of the seat coupling relative to the seat tube coupling.

The at least one motor may be fixed to one of the upper post and the lower post and may comprise a second drive shaft movably coupled to the linear actuator and operable to actuate the linear actuator and thereby cause linear translation of the upper post relative to the lower post.

The linear actuator may comprise a threaded or toothed lower shaft in rotatable threaded or toothed engagement with the other of the upper post and the lower post. The second drive shaft may be further operable to rotate the threaded or toothed lower shaft and thereby cause the upper post to translate relative to the lower post.

The linear actuator may further comprise a threaded ball nut fixed to the other of the lower post and the upper post. The threaded lower shaft may comprise a threaded ball screw rotatably engaging the threaded ball nut.

The linear actuator may comprise a roller nut fixed to the other of the lower post and the upper post, and the toothed lower shaft may comprise a roller screw rotatably engaging the roller nut.

The tilt controller may comprise a tilt control interface operable by the rider to actuate the tilt actuator when the bicycle is in motion.

The height controller may comprise a height control interface operable by the rider to actuate the linear actuator when the bicycle is in motion.

The tilt controller and the height controller may be integrated and may comprise a combined tilt and height control interface operable by the rider to actuate the tilt actuator and the linear actuator when the bicycle is in motion.

The prime mover may comprise a pressurized air chamber mounted in the lower post, the linear actuator may comprise a piston assembly with a piston chamber in the lower post fluidly coupled to the pressurized air chamber via an air valve, and a piston may be fixed to the upper post and movable within the lower post along the translation axis. The height controller may be further communicative with the air valve and operable to open the air valve to enable air to pass between the pressurized air chamber and the piston to thereby cause the upper post to translate linearly relative to the lower post.

The prime mover may further comprise a rotatable upper shaft comprising a drive gear. The tilt axis and upper shaft may be perpendicular to each other. The drive gear and seat coupling gear may be coupled to each other such that rotation of the upper shaft causes rotation of the seat coupling gear. The rotatable upper shaft may be fixed to the upper post along the translation axis and in rotatable engagement with the lower post, such that linear translation of the upper post relative to the lower post may cause the rotatable upper shaft to rotate.

The tilt actuator may further comprise a gear reduction unit rotatably coupling the rotatable upper shaft to the seat coupling gear.

The device may further comprise a resilient bias arranged to bias the upper post away from the lower post along the translation axis. The resilient bias may comprise a compression spring.

The device may further comprise a gear adjustment unit for adjusting an output of the first drive shaft relative to the second drive shaft.

The at least one motor may comprise a first motor movably coupled to the seat coupling gear, and a second motor movably coupled to the linear actuator.

The device may further comprise: a height sensor arranged to determine a height of the seat coupling relative to the seat tube coupling; a tilt sensor arranged to determine a tilt of the seat coupling relative to the seat tube coupling; and a control unit communicative with the height sensor and the tilt sensor and arranged, based on the determined height and tilt of the seat coupling relative to the seat tube coupling, to actuate one or more of the tilt actuator and the linear actuator. Actuating the tilt actuator may cause the first motor to be operated, and actuating the linear actuator may cause the second motor to be operated.

The control unit may be operable to cause the first motor to operate in response to operation of the second motor having caused the seat coupling to translate relative to the seat tube coupling. The control unit may be operable to cause the second motor to operate in response to operation of the first motor having caused the seat coupling to tilt relative to the seat tube coupling.

The device may further comprise a gear adjustment unit coupled to one or more of the first motor and the second motor. The gear adjustment unit may be configured to adjust an output of one or more of the first motor and the second motor.

The at least one motor may comprise at least one linear motor. The at least one linear motor may be coupled to a linear-rotary converter operable to convert a linear output of the at least one linear motor into torque for driving the seat coupling gear.

According to a further aspect of the disclosure, there is provided a device for adjusting a seat position of a bicycle seat, the device comprising: a seat tube coupling configured to couple to a bicycle seat tube; a seat coupling configured to couple to a bicycle seat; a seat adjustment mechanism comprising: a tilt actuator operable to adjust, about a tilt axis, a tilt of the seat coupling relative to the seat tube coupling; a linear actuator operable to linearly translate the seat tube coupling relative to the seat coupling; and a prime mover comprising at least one motor: movably coupled to the tilt actuator and operable to actuate the tilt actuator and thereby adjust, about the tilt axis, the tilt of the seat coupling relative to the seat tube coupling; and movably coupled to the linear actuator and operable to actuate the linear actuator and thereby cause linear translation of the seat tube coupling relative to the seat coupling.

The device may further comprise a tilt controller remote from and communicative with the seat adjustment mechanism and operable by a rider of a bicycle to actuate the tilt actuator and thereby adjust the tilt of the seat coupling relative to the seat tube coupling.

The device may further comprise a height controller remote from and communicative with the seat adjustment mechanism and operable by a rider of a bicycle to actuate the linear actuator and thereby adjust the height of the seat coupling relative to the seat tube coupling.

The tilt actuator may comprise a seat coupling gear fixed to the seat coupling. The at least one motor may be operable to drive rotation of a drive gear rotatably coupled to the seat coupling gear, and thereby cause rotation of the seat coupling gear.

One or more of the drive gear and the seat coupling gear may be bevelled, toothed, or may comprise one or more bearings.

The seat adjustment mechanism may further comprise: a lower post coupled to the seat tube coupling and defining a translation axis; and an upper post coupled to the seat coupling and translatable relative to the lower post along the translation axis. The linear actuator may be coupled to the lower post and upper post, and the linear actuator may be actuable by the prime mover to linearly translate the lower post relative to the upper post.

The at least one motor may be operable to drive a drive shaft movably coupled to the tilt actuator.

The at least one motor may be operable to drive a drive shaft movably coupled to the linear actuator.

The at least one motor may be fixed to one of the upper post and the lower post and may be operable to actuate the linear actuator and thereby cause linear translation of the upper post relative to the lower post. The linear actuator may comprise a threaded or toothed lower shaft in rotatable threaded or toothed engagement with the other of the upper post and the lower post, and the at least one motor may be further operable to rotate the threaded or toothed lower shaft and thereby cause the upper post to translate relative to the lower post. The linear actuator may further comprise a threaded ball nut fixed to the other of the lower post and the upper post and wherein the threaded lower shaft comprises a threaded ball screw rotatably engaging the threaded ball nut; or the linear actuator may further comprise a roller nut fixed to the other of the lower post and the upper post and wherein the toothed lower shaft comprises a roller screw rotatably engaging the roller nut.

The tilt controller may comprise a tilt control interface operable by the rider to actuate the tilt actuator when the bicycle is in motion.

The height controller may comprise a height control interface operable by the rider to actuate the linear actuator when the bicycle is in motion.

The tilt controller and the height controller may be integrated and may comprise a combined tilt and height control interface operable by the rider to actuate the tilt actuator and the linear actuator when the bicycle is in motion.

The seat adjustment mechanism may further comprise a gear adjustment unit for adjusting an output of the drive shaft movably coupled to the tilt actuator relative to the drive shaft movably coupled to the linear actuator.

The at least one motor may comprise a first motor movably coupled to the tilt actuator, and a second motor movably coupled to the linear actuator.

The at least one motor may comprise a motor movably coupled to the tilt actuator and movably coupled to the linear actuator.

The device may further comprise: a height sensor configured to determine a height of the seat coupling relative to the seat tube coupling; a tilt sensor configured to determine a tilt of the seat coupling relative to the seat tube coupling; and a control unit communicative with the height sensor and the tilt sensor and configured, based on the determined height and tilt of the seat coupling relative to the seat tube coupling, to actuate one or more of the tilt actuator and the linear actuator. Actuating the tilt actuator may comprise driving the drive shaft movably coupled to the tilt actuator, and actuating the linear actuator may comprise driving the drive shaft movably coupled to the linear actuator.

The control unit may be further configured to: drive the drive shaft movably coupled to the tilt actuator in response to translation of the seat tube coupling relative to the seat coupling; or drive the drive shaft movably coupled to the linear actuator in response to tilting of the seat coupling relative to the seat tube coupling.

The at least one motor may comprise at least one linear motor, and the at least one linear motor may be coupled to a linear-rotary converter operable to convert a linear output of the at least one linear motor into torque for driving one or more of the tilt actuator and the linear actuator.

According to further aspects of the disclosure, there are provided a bicycle comprising any of the above-described devices, and a kit of parts, comprising: a bicycle seat; and any of the above-described devices.

According to a further aspect of the disclosure, there is provided a device for adjusting a tilt of a bicycle seat, comprising: a seat post coupling for coupling to a bicycle seat post; a seat coupling for coupling to a bicycle seat; a seat tilt driver for rotating the seat coupling relative to the seat post coupling; one or more first sensors for determining an angle of the seat coupling relative to the seat post coupling; one or more second sensors for determining an angle of the seat post coupling relative to a reference position; and one or more processors configured to execute computer program code stored on a computer-readable medium to perform a seat tilt adjustment operation comprising: determining from the one or more first sensors the angle of the seat coupling relative to the seat post coupling; determining from the one or more second sensors the angle of the seat post coupling relative to the reference position; and adjusting, based on the determined angle of the seat coupling relative to the seat post coupling, and based on the determined angle of the seat post coupling relative to the reference position, the tilt of the bicycle seat by actuating the seat tilt driver.

The one or more second sensors may comprise one or more Global Navigation Satellite System (GNSS) sensors for determining a current position of the bicycle, and determining the angle of the seat post coupling relative to the reference position may comprise: determining from the one or more GNSS sensors a current geographic position of the bicycle; determining one or more slopes corresponding to the current geographic position of the bicycle; and determining, based on the one or more slopes, the angle of the seat post coupling relative to the reference position.

Determining the one or more slopes may comprise: accessing a database of geographic positions and corresponding slopes; and identifying in the database, based on the current geographic position of the bicycle, the one or more slopes.

Adjusting the tilt of the bicycle seat by actuating the seat tilt driver may comprise: accessing a database of seat tilt positions and corresponding bicycle inclinations; identifying in the database, using the determined angle of the seat post coupling relative to the reference position, one or more corresponding bicycle inclinations and one or more corresponding seat tilt positions; comparing the identified one or more seat tilt positions to the determined angle of the seat coupling relative to the seat post coupling; and adjusting, based on the comparison, the tilt of the bicycle seat by actuating the seat tilt driver.

Adjusting the tilt of the bicycle seat may comprise actuating the seat tilt driver to adjust the tilt of the bicycle seat toward the identified one or more seat tilt positions.

The device may further comprise one or more third sensors for determining a load applied to reference area. The seat tilt adjustment operation may further comprise: determining from the one or more third sensors the load applied to the reference area; and prior to adjusting the tilt of the bicycle seat, determining, using the determined load, whether to adjust the tilt of the bicycle seat by actuating the seat tilt driver.

The reference position may be a horizon.

The seat tilt driver may comprise one or more of: an electric motor; a stepper motor; an AC motor; and a DC motor.

The one or more first sensors may comprise one or more of: a magnetic sensor; a potentiometer; and an inertial sensor.

The device may further comprise one or more magnetic components fixed relative to one of the seat coupling and the seat post coupling, and the one or more first sensors may comprise one or more magnetic flux sensors fixed relative to the other of the seat coupling and the seat post coupling, for detecting one or more magnetic fields generated by the one or more magnetic components.

The one or more second sensors may comprise one or more of: an inertial sensor; and a magnetic sensor.

The one or more second sensors may comprise one or more gyroscopes for determining an angular velocity of the device in each of orthogonal x, y, and z axes, and one or more accelerometers for determining an acceleration of the device in each of the x, y, and z axes.

The device may further comprise a user input device. The computer-readable medium may further comprise computer program code configured when executed by the one or more processors to cause the one or more processors to perform a further seat tilt adjustment operation comprising: detecting one or more user inputs received at the user input device; and adjusting, based on the detected one or more user inputs, the tilt of the bicycle seat by actuating the seat tilt driver.

The device may further comprise: an adjustable-height seat post for coupling to the seat post coupling and comprising a seat tube coupling for coupling to a bicycle seat tube; a seat height driver for adjusting a height of the seat post coupling relative to the seat tube coupling; and a user input device. The computer-readable medium may further comprise computer program code configured when executed by the one or more processors to cause the one or more processors to perform a seat height adjustment operation comprising: detecting one or more user inputs received at the user input device; and adjusting, based on the detected one or more user inputs, the height of the seat post coupling relative to the seat tube coupling by actuating the seat height driver.

The user input device may comprise one or more of: an acoustic sensor for detecting a voice command; an optical sensor for detecting a gesture; and a manually operable control device.

The one or more second sensors may comprise one or more Global Navigation Satellite System (GNSS) sensors for determining a current position of the bicycle, and determining the angle of the seat post coupling relative to the reference position may comprise: determining from the one or more GNSS sensors a current position of the bicycle; accessing a database of geographic locations and corresponding slopes; identifying in the database, using the determined current position of the bicycle, one or more corresponding geographic locations and one or more corresponding slopes; and determining, using the identified one or more slopes, the angle of the seat post coupling relative to the reference position.

The database may be comprised in a topological map.

The geographic positions may comprise positions along one or more predetermined routes.

The device may further comprise one or more third sensors for determining a current position of the bicycle. The seat tilt adjustment operation may further comprise determining from the one or more third sensors the current position of the bicycle. Adjusting the tilt of the bicycle seat by actuating the seat tilt driver may be further based on the determined position of the bicycle.

The seat tilt driver may be configured to drive translation of a driving member coupled about a first pivot to a linkage system connected to the seat coupling such that translation of the driving member drives rotation of the seat coupling.

The linkage system may comprise a first linkage pivotally coupled about the first pivot to the driving member, and a linkage assembly connected to the seat coupling and pivotally coupled about a second pivot to the first linkage.

The linkage assembly may comprise a pair of second linkages, and the driving member may be operable translate between the pair of second linkages.

The linkage assembly may comprise a magnetic component for magnetically interacting with the one or more first sensors.

The one or more first sensors and the one or more second sensors may be comprised on one or more printed circuit boards (PCBs).

The one or more first sensors and the one or more second sensors may be comprised on one or more common PCBs.

The one or more GNSS sensors may be operable to receive data from or send data to one or more global satellite networks.

The device may further comprise a seat position driver for translating the seat coupling relative to the seat post coupling so as to thereby adjust a fore or aft position of the bicycle seat relative to the seat post coupling.

The seat position driver may comprise a motor operable to drive translation of the seat coupling relative to the seat post coupling by interaction with a rack fixed relative to the seat post coupling.

The one or more processors may be further configured to execute computer program code stored on a computer-readable medium to perform a seat position adjustment operation comprising: detecting a request to translate the seat coupling relative to the seat post coupling; and in response to detecting the request, actuating the seat position driver so as to translate the seat coupling relative to the seat post coupling.

According to a further aspect of the disclosure, there is provided a bicycle comprising a seat post and further comprising, attached to the seat post, a device for adjusting a tilt of a bicycle seat, according to any of the above-described embodiments.

The bicycle may further comprise one or more containers of pressurized gas stored within the seat post.

The seat post may comprise a lower seat tube translatable relative to an upper seat tube, and an anti-rotation mechanism for preventing rotation of the lower seat tube relative to the upper seat tube.

Each of the lower seat tube and the upper seat tube may define a longitudinal axis, and at least one of the lower seat tube and the upper seat tube may have a non-circular cross-section when taken perpendicularly to the longitudinal axis.

The non-circular cross-section may comprise a pair of linear portions and a pair of curved portions defining a periphery of the at least one of the lower seat tube and the upper seat tube.

The bicycle may further comprise a clamp for preventing the seat post from being decoupled from a frame of the bicycle. The clamp may comprise a clamp lock having a central recessed portion and multiple recessed lobe portions extending from the central recessed portion, at least one of the recessed lobe portions having a curved end.

The seat post may comprise a lower seat tube translatable relative to an upper seat tube in response to activation of a valve by a motor. The bicycle may further comprise a controller actuable by a rider of the bicycle when riding. The controller may be configured for wireless communication with the motor in order to wirelessly activate the valve.

According to a further aspect of the disclosure, there is provided a computer-readable medium storing computer program code configured when executed by one or more processors to cause the one or more processors to perform a seat tilt operation comprising: determining an angle of a seat coupling relative to a seat post coupling; determining an angle of the seat post coupling relative to a reference position; and causing, based on the angle of the seat coupling relative to the seat post coupling, and based on the angle of the seat post coupling relative to the reference position, a seat tilt driver to actuate so as to adjust a tilt of a bicycle seat.

The seat tilt operation may include any of the operations described above in connection with the device for adjusting a tilt of a bicycle seat.

This summary does not necessarily describe the entire scope of all aspects. Other aspects, features, and advantages will be apparent to those of ordinary skill in the art upon review of the following description of specific embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

Detailed embodiments of the disclosure will now be described in connection with the drawings, of which:

FIG. **1** is a cross-sectional view of a seat post assembly in accordance with a first, motorized embodiment of the disclosure;

FIG. **2** is a cross-sectional view of the seat post assembly of FIG. **1** in a lowered position;

FIG. **3** is an exploded view of an upper portion of the seat post assembly of FIGS. **1** and **2**;

FIG. **4** is an exploded view of a lower portion of the seat post assembly of FIGS. **1** and **2**;

FIG. **5** is a schematic representation of a bicycle incorporating a seat post assembly in accordance with an embodiment of the disclosure;

FIG. **6** is a flowchart showing a process of adjusting a position of a bicycle seat, in accordance with an embodiment of the disclosure;

FIG. **7** is a cross-sectional view of a seat post assembly in accordance with a second, mechanized embodiment of the disclosure;

FIG. **8** is a cross-sectional view of the seat post assembly of FIG. **7** in a lowered position; and

FIG. **9** is a cross-sectional view of an alternative arrangement of the seat post assembly of FIG. **1**.

FIG. **10**A shows a tilt adjustment device attached to a seat post, according to an embodiment of the disclosure.

FIG. **10**B shows a tilt adjustment device according to an embodiment of the

disclosure.

FIGS. **11**A-**12**C show different views of an interior of the tilt adjustment device of FIG. **10**A, according to an embodiment of the disclosure.

FIG. **13** is a flow diagram of a method of determining an inclination of a bicycle, according to an embodiment of the disclosure.

FIG. **14** is a flow diagram of a method of adjusting a tilt of a bicycle saddle, according to an embodiment of the disclosure.

FIG. **15** is a flow diagram of a method of determining an inclination of a bicycle, according to an embodiment of the disclosure.

FIG. **16** shows a bicycle saddle at different tilt angles, according to an embodiment of the disclosure.

FIG. **17** shows an inclination angle of a slope, according to an embodiment of the disclosure.

FIG. **18** shows a lower seat tube within an upper seat tube, as part of a seat post for a bicycle, according to an embodiment of the disclosure.

FIGS. **19**A and **19**B show a seat clamp according to an embodiment of the

disclosure.

FIG. **20** shows a cross-section of a valve assembly according to an embodiment of the disclosure.

FIG. **21** shows a tilt adjustment device configured for translation in the fore and aft directions, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

The present disclosure seeks to provide an improved device for adjusting a seat position of a bicycle seat. Whilst various embodiments of the disclosure are described below, the disclosure is not limited to these embodiments, and variations of these embodiments may well fall within the scope of the disclosure which is to be limited only by the appended claims.

A brief summary of embodiments of the disclosure follows. This summary is not to be seen as limiting in any way on the scope of the disclosure. Generally, there is described a device for adjusting a position of a bicycle seat. The device includes an upper post that may translate relative to a lower post. The upper post includes a seat coupling which is arranged to couple to a bicycle seat. The lower post includes a seat tube coupling which allows the device to be coupled to a bicycle seat tube. The device also includes a seat adjustment mechanism. The seat adjustment mechanism, by driving movement of the seat coupling relative to the seat tube coupling, is used to adjust a tilt and/or a height of a bicycle seat coupled to the seat coupling. A tilt and height controller is provided on the handlebars of the bicycle and may be controlled by a rider of the bicycle during movement of the bicycle (i.e. when the rider is pedalling). Actuation of the tilt and height controller operates a prime mover which is responsible for providing the force necessary to drive a tilt actuator and a linear actuator which form part of the seat adjustment mechanism.

Two different embodiments of the disclosure are described: a motorized embodiment and a mechanized embodiment. In the motorized embodiment, the prime mover includes an electrical motor. The motor drives a tilt actuator, which forms part of the seat adjustment mechanism and comprises a bevelled seat coupling gear fixed to the seat coupling. Operation of the motor drives rotation of a bevelled drive gear which is in a bevelled coupling with the bevelled seat coupling gear. Thus, operation of the motor converts rotation of the drive gear about the z-axis (i.e. the axis along which the upper and lower posts translate) into rotation of the seat coupling gear about the x-axis (i.e. the axis about which the seat coupling rotates or tilts). Rotation of the seat coupling gear about the x-axis results in tilting of the seat coupling and corresponding tilting of the bicycle seat.

In an alternative embodiment, the electrical motor (or at least one of the electrical motors) may be configured to output a linear force, as opposed to a torque. Thus, operation of the electrical motor (or one of multiple electrical motors) drives a change in linear position of the motor's drive shaft. A linear-rotary converter then converts the change in linear position of the drive shaft into rotary motion which acts upon the bevelled drive gear which is in a bevelled coupling with the bevelled seat coupling gear, thereby adjusting seat tilt as described above.

Operation of the motor is further arranged to drive a linear actuator, which forms part of the seat adjustment mechanism. The linear actuator comprises a threaded lower shaft in rotatable threaded engagement with the lower post. The lower shaft is fixed to the upper post, and thus rotation of the lower shaft is converted into translation of the upper post relative to the lower post. Operation of the linear actuator therefore drives translation of the upper post relative to the lower post.

In the motorized embodiment, actuation of the motor (i.e. the prime mover) is initiated using the tilt and height controller positioned on the bicycle's handlebars. The controller is communicative with the motor and operates rotation of the linear actuator, resulting in a change in height of the bicycle seat. The change in height is detected by a sensor which is in communication with a control unit. The control unit communicates in turn with the motor to drive operation of the tilt actuator, thereby adjusting a tilt angle of the bicycle seat as a function of the height adjustment of the bicycle seat. The device therefore allows for automatic adjustment of the saddle tilt in conjunction with, and as a function of, adjustment of the saddle height.

According to some embodiments, instead of adjusting tilt in response to a detected change height, the seat adjustment mechanism may be configured such that height is adjusted in response to a detected change in tilt. Furthermore, according to some embodiments, a gear adjustment unit may be used such that rotation of one of the motor's drive shafts is converted into corresponding rotation of another of the motor's drive shafts (i.e. rather than having two independently controlled drive shafts). One of the drive shafts is configured to drive the linear actuator, and the other of the drive shafts is configured to drive the tilt actuator.

In the mechanized embodiment, the prime mover comprises a source of pressurized air, and the linear actuator comprises a piston assembly. The rider may use the height and tilt controller to initiate release of the pressurized air. The pressurized air operates the linear actuator by urging the piston assembly to move along the translation axis. As the piston assembly is fixed to the upper post, operation of the linear actuator drives translation of the upper post relative to the lower post. Thus, using the controller allows the upper post to move both upwards relative to the lower post (if the rider removes sufficient weight from the saddle) and also downwards relative to the lower post (if the rider applies sufficient weight to the saddle).

Similarly to the motorized embodiment, the mechanized embodiment includes a threaded upper shaft fixed to the upper post and in threaded engagement with the lower post. Thus, translation of the piston assembly results in translation of the threaded upper shaft relative to the lower post. Due to the threaded engagement of the threaded upper shaft with the lower post, the threaded upper shaft is caused to rotate about the translation axis as it moves along the translation axis. The mechanized embodiment uses the same tilt actuator as the motorized embodiment. In particular, the threaded upper shaft is rotatably coupled to the drive gear such that rotation of the threaded upper shaft results in corresponding rotation of the seat coupling gear about the tilt axis. Thus, as the upper post translates relative to the lower post, the seat coupling is simultaneously rotated about the tilt axis, resulting in tilting of the bicycle seat. The device therefore allows for adjustment of the saddle height simultaneously to adjustment of the saddle tilt.

A detailed description of these two embodiments will now follow. A complete list of parts referenced in the drawings is included at the end of the description. However, where it is considered that a full description of any of the parts would not assist a person skilled in the art in understanding the disclosure, a description of the part in question has been omitted.

In accordance with a first embodiment of the disclosure, FIGS. **1** and **2** show a seat post assembly **1000** configured to adjust both a height and a tilt of a bicycle saddle. As will be described in more detail below, this embodiment uses motorized means for adjusting the height and tilt of the saddle, and may therefore be referred to as the motorized embodiment. FIG. **1** shows seat post assembly **1000** in a raised position, whereas FIG. **2** shows seat post assembly **1000** in a lowered position. In FIGS. **1** and **2**, like elements are numbered using like reference numbers. The description begins with a description of seat post assembly **1000** in the raised position (FIG. **1**).

Seat post assembly **1000** comprises a cylindrical upper post **10** translatable along the z-axis within a cylindrical lower post **12**. Upper post **10** is therefore arranged telescope in and out of lower post **12**. For the purposes of this description, the z-axis is held to be the axis along which upper post **10** and lower post **12** translate relative to each other. Lower post **12** may accommodate one of various standardized seat post diameters, such as standard to oversized and even greater. Of course, lower post **12** may have any other diameter suitable for its use with a bicycle.

The upper end of lower post **12** is coupled via a collar **14** to upper post **10**. At this coupling is provided a dust seal wiper **16** for preventing dust and other contaminants from entering the coupling of lower post **12** to upper post **10**. Below collar **14** are provided alignment keys **18** to ensure that, during telescoping/translating, upper post **10** does not spin or rotate within lower post **12**. Lower post **12** is affixed at its base to a bicycle's seat tube (not shown). As known in the art, the bicycle seat tube is a permanent element of a bicycle frame which is used to hold and secure in place a bicycle seat post.

Within lower post **12** is a cylindrical post baffle **20**. The position of post baffle **20** within lower post **12** defines an annular channel **22** which provides a space to allow upper post **10** to telescope within lower post **12**. The base of post baffle **20** is secured to lower post **12** via a threaded and ported coupler insert **24**. Coupler insert **24** threads or inserts snuggly into lower post **12** and defines a base for post baffle **20**. Coupler insert **24** is fastened with a body circlip **26** that clicks into place to prevent lateral movement or backing off of coupler insert **24** from lower post **12**.

The lower portion of lower post **12** forms, beneath coupler insert **24**, a chamber **28**. Chamber **28** is used to house a number of electronic components. These electronic components include a control unit **30**, a mounting sleeve **32**, a driver **34**, a lower battery mount **38**, a battery **40**, an upper battery mount **42** and a charging/data port **44**. The base of lower post **12** is sealed to prevent the ingress of contaminants into chamber **28**. To this end a threaded bottom cap **46** engages the bottom of lower post **12** by threading internally into the bottom opening of lower post **12**. An additional hydrophobic coating may be applied to chamber **28** and the electrical components within chamber **28** to further prevent the possibility of a short circuit.

Above lower battery mount **38** is housed battery **40**. Battery **40** is sealed to prevent damage to its circuitry/components. Battery **40** is held in place from above via upper battery mount **42** which provides reinforced insulation to the upper side of battery **40**. Through a number of leads and other electrical connections, battery **40** provides power to various electrical components in chamber **28**, including control unit **30** and driver **34**. Charging/data port **44** may be connected to an external power source for recharging battery **40**, and/or may be connected to a PC or other computing device for data management purposes. In other embodiments (not shown), the battery may be located within the seat tube of the bicycle frame.

A channel **50** is formed within coupler insert **24** and provides a space through which an electrical conduit **48** may pass through channel **50**. Electrical conduit **48** carries electrical wiring from control unit **30**, driver **34** and battery **40** to electrical motor **74** described in further detail below. A conduit bushing **52** extends from a wiring harness **118**, through channel **50** and through electrical conduit **48**. Conduit bushing **52** fully encases the wiring between wiring harness **118** and a guide plate **156** (see below).

Above chamber **28**, along the z-axis, there is provided a dual ported linear motion piston **54** configured to translate along the z-axis in correspondence with movement of upper post **10** within lower post **12**. Piston **56** is coupled to a high helix lead ball screw **58** via a lead screw termination mount **70**. Screw **58** is configured to rotate about the z-axis through activation of motor **74**, as will be described in more detail below. A fixed piston guide **60** is secured to the top of post baffle **20**. Piston guide **60** is configured to guide rotation of screw **58** within the threads of an affixed lead ball nut **62**. The carriage of nut **62** encases the arbour of screw **58**, and nut is fixed to lower post **12**. Thus, movement of screw **58** linearly along the z-axis will cause corresponding rotation of screw **58** around the z-axis through its threaded engagement with nut **62**. A recoil bellow **64** encases screw **58** for protection. A piston cavity **66** exists between coupler insert **24** and a threaded and dual ported lead guide plate **156**, and is bounded in part by post baffle **20**. Piston **54** is configured to move within piston cavity **66** during translation of piston **54** along the z-axis. Affixed to piston **54** is a height inertial sensor **68**. Height inertial sensor **68** is arranged to detect an amount of translation of upper post **10** relative to lower post **12**.

A rigid housing ferrule **94** adapts the internal wiring from conduit bushing **52** into a recoil housing **96**. This arrangement ensures the wiring can be coiled for optimum length around the circumference of upper post **10**. The arrangement furthermore prevents telescoping of upper post **10** in/out of lower post **12** from interfering with the wiring.

Now turning our attention to the top of seat post assembly **1000**, there is provided at the top of upper post **10** a bulkhead assembly **72**. Beneath bulkhead assembly **72** and housed within upper post **10** is an electric motor **74**. Motor **74** may be a D/C motor, an A/C motor, a stepper motor, a gear motor, a stack motor, a gear head motor, a linear motor, a brushless motor, a hysteresis motor, a reluctance motor, a universal motor, a piezoelectric motor, a magnetic motor, a pneumatic motor, a hydraulic motor, or any other suitable type of electric or non-electric motor that may be used to implement the seat position adjustment method described herein. Motor **74** is a dual-head motor and rotates two heads about the z-axis. The dual heads of motor **74** include a lower motor head **76** which faces downward (toward chamber **28**) along the z-axis, and an upper motor head **78**, which faces upward (toward bulkhead assembly **72**) along the z-axis. Both lower motor head **76** and upper motor head **78** may rotate in both clockwise and counter-clockwise directions, when actuated by a controller **90** (not shown in FIG. **1**). Lower motor head **76** is coupled to screw **58** via a number of fastening components (referenced in the drawings but not described in more detail here). These components assist in driving a smooth transmission from rotation of lower motor head **76** to rotation of screw **58**. Due to the coupling of lower motor head **76** and screw **58**, any rotation of lower motor head **76** by motor **74** will result in a corresponding rotation of screw **58**.

Upper motor head **78** is coupled to a bevel gear mechanism **80** via a number of fastening components (referenced in the drawings but not described in more detail here). These components assist in driving a smooth transmission from rotation of upper motor head **78** to rotation of bevel gear mechanism **80**. Coupled to bevel gear mechanism **80** is a pair of splined satellite bevel gear assemblies **82** in turn coupled to a pair of rail clamp assemblies **84**. Due to the bevelled coupling of bevel gear assemblies **82** to bevel gear mechanism **80**, bevel gear assemblies **82** are configured to rotate about an axis perpendicular to the rotation axis of bevel gear mechanism **80**. In other words, bevel gear assemblies **82** are configured to rotate about an axis perpendicular to the z-axis (i.e. the x-axis). Through the coupling of bevel gear assemblies **82** to rail clamp assemblies **84**, rotation of bevel gear assemblies **82** about the x-axis results in corresponding rotation of rail clamp assemblies **84** about the x-axis. Affixed onboard bevel gear mechanism **80** is a tilt inertial sensor **86**. Tilt inertial sensor **86** is arranged to detect an amount of rotation of bevel gear mechanism **80**, and therefore an amount of rotation of a saddle when coupled to bulkhead assembly **72**. As known in the art, rail clamp assemblies **84** are configured to clamp or otherwise secure a bicycle saddle relative to seat post assembly **1000**. Rail clamp assemblies **84** are configured to accommodate various rail-clamp diameters including standard and oversized twin rail clamps, and may include single post or beam-like clamping mechanisms.

According to some embodiments, the motor may comprise a linear or stepper-like motor. In order to convert the linear output of such a motor into driving and holding torque, a dogbone-like linkage system, or other linear-rotary converter, may be used to couple the motor's output to bevel gear assemblies **82**. Such a motor and associated linear-rotary converter, at the cost of increased weight and a greater number of moving parts, may provide a potentially higher threshold holding torque, and may reduce the need for a step-up torque/gear assembly coupled to lower motor head **76**. An illustration of such an embodiment is shown in FIG. **9**.

For additional detail, an exploded view of seat post assembly **1000** is shown in FIGS. **4** and **5**. Like elements are numbered using like reference numbers.

In use, seat post assembly **1000** functions as follows. Seat post assembly **1000** is mounted to a bicycle frame of a bicycle **88**, for example as schematically shown in FIG. **3**. Bicycle **88** includes a controller **90** located on the handlebars of bicycle **88** and therefore within easy reach of a rider. Controller **90** is electrically connected to motor **74**.

While riding bicycle **88**, the rider may desire to readjust the height and/or the tilt of the saddle **92**. For example, if approaching a steep downhill section, the rider may wish to lower the height of saddle **92** as well as tilt saddle **92** upwards. Whilst still in motion, the rider activates controller **90** to initiate a seat adjustment. Activating controller **90** triggers operation of motor **74**. The signal received at motor **74** causes motor **74** to rotate lower motor head **76**.

Rotation of lower motor head **76** is transmitted to screw **58**. Because of the engagement of the threads of screw **58** with nut **62**, rotation of screw **58** is converted into linear motion of nut **62** along the z-axis. Because nut **62** is fixed to lower post **12**, upper post **10** is caused to telescope relative to lower post **10**. In particular, upper post **10** (coupled to screw **58**) is caused to telescope into lower post **12** (coupled to nut **62**). The linear motion of upper post **10** is guided by piston **54**. As a result, the z-position or height of bulkhead assembly **72** relative to the bicycle frame will decrease.

As upper post **10** telescopes within lower post **12**, height inertial sensor **68** detects the amount of translation of upper post **10** relative to lower post **12**. Height inertial sensor **68** is in communication (wired or otherwise) with control unit **30** and provides as, an input to control unit **30**, data regarding the amount of height adjustment (i.e. the degree of translation of upper post **10** relative to lower post **12**). Based on this input, control unit **30** determines by how much the tilt of the saddle is to be adjusted (as a function of by how much the height of the saddle has been adjusted). Control unit **30** instructs motor **74** to operate upper motor head **78** so as to raise the tilt of saddle **90**. A gear reduction unit **148** within motor **74** provides a predetermined reduction in the rotation of upper motor head **78** relative to the rotation of lower motor head **76**. Through the bevelled engagement of bevel gear mechanism **80** with bevel gear assemblies **82**, rotation of upper motor head **78** results in rotation of rail clamp assemblies **84** about the x-axis. Rotation of rail clamp assemblies **84** about the x-axis axis results in an adjustment of a tilt angle of saddle **92** relative to the horizontal. In particular, in the present example of the rider approaching a downhill section and activating controller **92**, saddle height is decreased and saddle **90** is tilted upwards.

In embodiments where tilt is adjusted in response to height, a step-up torque/gear assembly may be included in order to provide a predetermined increase in the rotation of upper motor head **78** relative to the rotation of lower motor head **76**.

In embodiments where multiple motors provide independently controllable outputs to both the seat tilt adjustment and the seat height adjustment, there may be no need for a gear reduction unit or a gear increase unit. However, a gear reduction unit or a gear increase unit may be used to provide fine-tuned position adjustments to the seat height/seat tilt.

It is generally desirable to tilt a saddle upwards and lower saddle height for downhill sections. Similarly it is generally desirable to tilt a saddle downwardly and raise saddle height for uphill sections. Therefore the respective rotation directions of lower motor head **76** and upper motor head **78** may be pre-set accordingly. In addition the ratio of saddle tilt adjustment to saddle height adjustment may be pre-set, or alternatively may be adjusted at any point through appropriate adjustment of gear reduction mechanism **148**. Generally, it is desirable that 16 mm change in height corresponds to 0.4° to 2.5° of tilt, although the particular preference will vary from one rider to the next.

FIG. **2** shows seat post assembly **1000** in the lowered position. Of course, upper post **10** may be telescoped back out of lower post **12** by activating controller **90** again such that the lower and upper motor heads **76**, **78** rotate in opposite directions to that in which they rotated when lowering the seat height.

Thus, rotation of lower motor head **76** in either of two directions results in corresponding lowering or raising of bulkhead assembly **72**, through the threaded engagement of screw **58** with nut **62** fixedly coupled to lower post **12**. The translation of upper post **10** relative to lower post **12** is detected by height inertial sensor **68** and communicated to control unit **30**. Control unit **30** instructs near simultaneous rotation of upper motor head **76** to provide corresponding rotation of rail clamp assemblies **84** about the x-axis, thereby tilting a saddle attached to bulkhead assembly **72** either upwards or downwards (depending on the direction of translation of upper post **10**).

It is conceivable that a rider may wish to operate lowering/raising of the saddle independently of saddle tilt. Thus, controller **90** may be configured to provide control of lower motor head **76** independently of upper motor head **78**, and vice versa. In this case, seat post assembly **1000** may be provided without inertial sensors **68** and **86**, and without control unit **30**, and controller **90** would then operate each motor head **76**, **78** independently of the other. The rider could therefore be provided with three seat position adjustment options via controller **90**: 1) adjust seat height and seat tilt simultaneously; 2) adjust seat height only; and 3) adjust seat tilt only. Alternatively, it is envisaged that motor **74** could be replaced with two individual motors, each motor operating one of lower motor head **76** and upper motor head **78** independently of the other. In this embodiment, controller **90** would include a controller for operating one motor, and another controller for operating the second motor.

FIG. **6** illustrates an exemplary process **600** that may be executed by control unit **30** in order to effect automatic seat height adjustment and seat tilt adjustment. At step **610**, process **600** commences. At step **620**, in response to a change in height of the saddle, the new height position of the saddle is received at control unit **30**. In particular, the new height position is determined by height inertial sensor **68** and sent to control unit **30**. At step **630**, the current tilt position of the saddle is received. In particular, the current tilt position is determined by tilt inertial sensor **86** and sent to control unit **30**. In other embodiments, rather than the inertial sensors sending their respective position information to control unit **30**, control unit **30** may query height inertial sensor **68** and tilt inertial sensor **86** for height and tilt position information, respectively. At step **640**, control unit **30** compares the current tilt position to the new height position. The comparison may include a determination of whether, for a given height position, the tilt position is within a pre-set range tilt positions. The present ranges may be stored in a database of pre-set ranges, stored within a memory of control unit **30**.

At step **650**, control unit **30** determines whether the current tilt position is optimal, based on the comparison at step **630**. An optimal tilt position may be a tilt position which is, for a given height position, within a pre-set range tilt positions. If the tilt position is outside of the pre-set range, then control unit **30** determines that the current tilt position is not optimal. In this case, control unit **30** sends an instruction to motor **74** to initiate operation of the tilt position adjustment mechanism. In other words, control unit **30** causes motor **74** to initiate rotation of upper motor head **78** so as to adjust the tilt position of the saddle. The tilt position of the saddle is then adjusted as a function of the new height position of the saddle. Thus, during use of seat post assembly **1000**, adjustment of the height of the saddle (i.e. adjustment of the height position) will lead to an automatic and corresponding adjustment of a tilt of the saddle (i.e. adjustment of the tilt position).

In alternative embodiments, it is envisaged that, instead of tilt position being adjusted in response to height position, it is height position which is adjusted in response to a change in tilt position. Thus, actuation of controller **90** will cause operation of the tilt adjustment mechanism. Operation of the tilt adjustment mechanism results in a change of the saddle's tilt position which is measured by tilt inertial sensor **86**. Tilt inertial sensor **86** communicates the new tilt position to control unit **30**. Control unit **30** then compares the new tilt position to the current height position and determines whether the current height position is no longer optimum relative to the new tilt position. If the current height position is no longer optimum, then control unit **30** then sends an instruction to motor **74** to operate the height adjustment mechanism (i.e. by causing rotation of screw **58**) to cause upper post **10** to translate relative to lower post **12**.

The speed at which control unit **30** operates is sufficiently high such that, to the rider, the tilt adjustment mechanism and the height adjustment mechanism operate substantially simultaneously, although in reality one is operated in response to operation of the other.

In accordance with a second embodiment of the disclosure, FIGS. **8** and **9** show a seat post assembly **2000** configured to adjust both a height and a tilt of a bicycle saddle. This embodiment uses only mechanical means for adjusting the height and tilt of the saddle, and is therefore referred to as the mechanized embodiment. FIG. **7** shows seat post assembly **2000** in a raised position, whereas FIG. **8** shows seat post assembly **2000** in a lowered position. In FIGS. **8** and **9**, like elements are numbered using like reference numbers. The description begins with a description of seat post assembly **2000** in the raised position (FIG. **8**).

Seat post assembly **2000** comprises a cylindrical upper post **11** translatable along the z-axis within a cylindrical lower post **13**. Upper post **11** is therefore arranged telescope in and out of lower post **13**. Lower post **13** may accommodate one of various standardized seat post diameters, such as standard to oversized and even greater. Of course, lower post **13** may have any other diameter suitable for its use with a bicycle.

The upper end of lower post **13** is coupled via a collar **15** to upper post **11**. At this coupling is provided a dust seal wiper **17** for preventing dust and other contaminants from entering the coupling of lower post **13** to upper post **11**. Below collar **15** are provided alignment keys **19** to ensure that, during telescoping/translating, upper post **11** does not spin or rotate within lower post **13**. Lower post **13** is affixed at its base to a bicycle's seat tube (not shown). As known in the art, the bicycle seat tube is a permanent element of a bicycle frame which is used to hold and secure in place a bicycle seat post.

In the base of lower post **13** is provided a base inflation valve **21**. Base inflation valve **21**, when open, provides a fluid communication flow path from the exterior of seat post assembly **2000** to an air chamber **23**. Air can be pumped into chamber **23** via base inflation valve **21** in order to provide a source of pressurised air within chamber **23**. The valve stem on base inflation valve **21** may be a Schrader valve stem, a Presta valve stem, or any other suitable valve stem which would require a pumping tool such as a bicycle pump to obtain the desired pressure within chamber **23**. O-rings **25**, **27** and a threaded bottom cap **29** are used to seal chamber **23**.

A controller **31** extends from within chamber **23** to a location where controller **31** may be operated by the bicycle rider while the bicycle is in motion. Controller **31** may be a lever, a button, or other like device, and typically is mounted on the handlebars of the bicycle, for easy reach by the rider. In the embodiment of FIG. **8**, controller **31** is a pressurized hydraulic oil controller.

Chamber **23** houses a pressurized air delivery system configured to allow pressurized air to flow from within chamber **23** to piston cavity **33**. In particular chamber **23** houses, amongst other components, a valve ball **35** and a valve seat **37**. Various other components are housed within chamber **23**, such as a spool valve. These are not described in detail here but are referenced accordingly in FIG. **1**.

When controller **31** is actuated via a wirelessly configured electrical actuator (e.g. a piezoelectric motor) communicative via Bluetooth™, Zigbee™, Z-Wave™, ANT+, or Wi-Fi) , hydraulic pressure is controlled to displace the position of valve ball **35** relative to valve seat **37**. A fluid flow path is then opened from chamber **23** to piston cavity **33**, and pressurized air may flow from chamber **23** into piston cavity **33** via an air aperture **39** at the base of piston cavity **33**. A threaded orifice coupler insert **41** is provided between chamber **23** and piston cavity **33** and serves to provide a secure seal between chamber **23** and piston cavity **33**.

Above chamber **23** and along the z-axis is located a high helix lead ball screw, rod, or rail **43** joined at one end to a centrally ported linear motion piston **45**. Piston **45** is configured to translate vertically along the z-axis relative to lower post **13**. High tolerance rod O-rings **47**, **49** prevent air from escaping from piston cavity **33**, as piston **45** moves dynamically or remains static. Therefore, air may only enter and exit piston cavity **33** via air aperture **39** connecting chamber **23** and piston cavity **33**. A compression spring **51** is housed within upper post **11** and arranged to exert a downwardly biasing force on piston **45**. That is, spring **51** is configured to urge upper post **11** to telescope out of lower post **13**.

An affixed lead ball nut **53** is affixed to lower post **13**. The carriage of nut **53** encases the arbour of screw **43**. Thus, movement of piston **45** along the z-axis and within lower post **13** will urge screw **43** to move linearly within nut **53**. Because nut **53** is fixed to lower post **13**, nut **53** imparts rotational motion of screw **43** during translation of screw **43**. Thus, movement of piston **45** within lower post **13** results in rotation of screw **43** about the z-axis. Rotation of screw **43** is guided via a mid-lead guide bearing **55** and a threaded and ported lead guide plate **57**.

At the topmost end of screw **43**, screw **43** enters a gear reduction unit **59**. Gear reduction unit **59** is dual-headed and comprises a lower gear reduction driver **61** which faces downward (toward piston **45**) and an upper input driver **63** which faces upward (away from piston **54**). Both lower gear reduction driver **61** and upper input driver **63** may rotate in both clockwise and counter-clockwise directions. Lower gear reduction driver **61** is coupled to screw **43** via a number of fastening components (referenced in the drawings but not described in more detail here). These components assist in driving a smooth transmission from rotation of lower gear reduction driver **61** to rotation of screw **45**. Upper input driver **63** is coupled to a bevel gear mechanism **65** via a number of fastening components (referenced in the drawings but not described in more detail here). These components assist in driving a smooth transmission from rotation of upper input driver **63** to rotation of bevel gear mechanism **65**.

Coupled to bevel gear mechanism **65** is a pair of splined satellite bevel gear assemblies **67** in turn coupled to a pair of rail clamp assemblies **69**. Due to the bevelled coupling of bevel gear assemblies **67** to bevel gear mechanism **65**, bevel gear assemblies **67** are configured to rotate about an axis perpendicular to the rotation axis of bevel gear mechanism **5**. In other words, bevel gear assemblies **7** are configured to rotate about an axis perpendicular to the z-axis (i.e. the x-axis). Through the coupling of bevel gear assemblies **67** to rail clamp assemblies **69**, rotation of bevel gear assemblies **67** about the x-axis results in corresponding rotation of rail clamp assemblies **69** about the x-axis. As known in the art, rail clamp assemblies **69** are configured to clamp or otherwise secure a bicycle saddle relative to seat post assembly **2000**.

In use, seat post assembly **2000** functions as follows. Seat post assembly **2000** is mounted to a bicycle frame of a bicycle (for example a bicycle as shown in FIG. **5**). Controller **31** is on the handlebars of the bicycle and therefore within easy reach of the rider. While riding the bicycle, the rider may desire to readjust the height and/or the tilt of the saddle. For example, if approaching a steep downhill section, the rider may wish to lower the height of the saddle as well as tilt the saddle upwards. Whilst still in motion, the rider activates controller **31** to initiate a seat adjustment.

Activating controller **31** causes valve ball **35** to move away from valve seat **37**, as described above. When the valve is opened, the pressurized air within air chamber **23** flows into piston cavity **33** via aperture **39**. The expansion of the air as it flows into piston cavity **33** results on an upwards force being exerted on piston **45**. Piston **45** is therefore urged upwards by the pressurized air entering piston cavity **33**. However, with the rider's full weight applied on the saddle, the weight is sufficient to overcome the upward force exerted on piston **45**. As a result piston **45** will translate downwards, compressing the air in piston cavity **33** back into chamber **23** through aperture **39**. As piston **45** is coupled to upper post **11**, upper post **11** will translate downwards by telescoping into lower post **13**. Compression spring **51** ensures that the downward translation of piston **45** is sufficiently dampened to prevent the rider from completely telescoping upper post **11** within lower post **13** and potentially damaging the device.

Once upper post **11** has reached the desired (lower) height, the rider may deactivate controller **31**. Deactivation of controller **31** results in valve ball **35** realigning with valve seat **37** and preventing air from flowing between piston cavity **33** chamber **23**. Once the valve is closed, the upward force exerted on piston **45** by the compressed air in piston cavity **33** balances the downward force exerted on piston **45** by the rider's weight. The rider may then reapply their full weight on the saddle as, even with their full weight applied, it is insufficient to further compress the air contained within piston cavity **33**.

Because of the engagement of the threads of screw **43** with stationary nut **53**, translation of upper post **11** along the z-axis results in rotation of screw **43** about the z-axis. Rotation is screw **43** is transmitted to lower gear reduction driver **61**, input bevel gear mechanism **63** and bevel gear mechanism **65**. Through the bevelled engagement of bevel gear mechanism **65** with bevel gear assemblies **67**, rotation of input bevel gear mechanism **63** results in rotation of rail clamp assemblies **69** about the x-axis. Rotation of rail clamp assemblies **69** about the x-axis axis results in an adjustment of a tilt angle of the bicycle saddle relative to the horizontal. In particular, in the present example of the rider approaching a downhill section and activating controller **31**, saddle height is decreased and the saddle is tilted upwards. FIG. **8** shows seat post assembly **2000** in a lowered position.

Conversely, the rider may wish to raise their saddle from a lowered position to a raised position. The rider would then activate controller **31**, thereby opening the valve and releasing pressurized air into piston cavity **33**. The release of pressurized air urges piston **45** upwards. In order to allow piston **45** and upper post **11** to translate upwards, the rider would need to lift some of their weight off the saddle, for example by raising their hips slightly. When sufficient weight has been removed, piston **45** will translate upwards under the force of the pressurized air expanding into piston cavity **33**. Compression spring **51** assists with the upward translation of piston **45**. Once the desired height is reached, the rider reactivates controller **31** to close the valve. Again, during upwards translation of piston **45**, screw **43** will be caused to rotate through its threaded engagement with nut **53** which is affixed to lower post **13**. The rotation of screw **43** results in tilting of the saddle through bevel gear mechanism **65** and bevel gear assemblies **67** as described above (in this case, during raising of the saddle, the saddle tilts downwards).

Thus, vertical translation of piston **45** results in lowering or raising of the saddle, through the action of pressurized air entering piston cavity **33**. Simultaneously, translation of screw **43** through threaded nut **53** results in rotation of screw **43**. Rotation of screw **43** causes bevel gear mechanism **65** and bevel gear assemblies **67** to tilt the saddle either upwards or downwards, depending on which way screw **43** is rotating.

Furthermore, it is conceivable that, given the mechanized embodiment, a rider may wish to operate lowering/raising of the saddle independently of saddle tilt. The mechanized embodiment could be modified such the rotation of the threaded screw **43** is decoupled from the bevel gear mechanism **65**. The bevel gear mechanism could then be operated using a second controller positioned for example on the handlebars. For instance, a rack and pinion-type arrangement could be used to convert linear motion of the controller into rotational motion arranged to rotate the bevel gear mechanism. This is merely once possible example of how, in the mechanized embodiment, lowering/raising of the saddle could be operated independently of saddle tilt.

PARTS LIST-MOTORIZED EMBODIMENT

10 Upper post
12 Lower post
14 Collar
16 Dust seal wiper
18 Alignment keys
20 Post baffle
22 Annular channel
24 Coupler insert
26 Body circlip
28 Chamber
30 Control unit
32 Mounting sleeve
34 Driver
36 Detent
38 Lower battery mount
40 Battery
42 Upper battery mount
44 Charging/data port
46 Threaded bottom cap
48 Electrical conduit
50 Channel
52 Conduit bushing
54 Piston
58 Screw/shaft
60 Piston guide
62 Ball nut/roller
64 Recoil bellow
66 Piston cavity
68 Height inertial sensor
70 Lead screw termination mount
72 Bulkhead assembly
74 Electric motor
76 Lower motor head
78 Upper motor head
80 Bevel gear mechanism
82 Bevel gear assemblies
84 Rail clamp assemblies
86 Tilt inertial sensor
88 Bicycle
90 Controller
92 Saddle
94 Housing ferrule
96 Recoil housing
98 O-ring
100 Bottom cap spring
102 Threaded bottom cap
104 Body circlip
106 O-ring
108 Flat washer
110 Battery leads
112 Battery lead connectors
114 Driver leads
116 Control unit leads
118 Wiring harness
120 O-ring
122 Lower lead guide bearing
124 Mid lead guide bearing
126 Upper bellow washer
128 Upper lead cavity
130 Lower threaded motor coupler mount
132 Upper motor dampening spacer
134 Input driver
136 Exterior rail clamp
138 Hex rail fastener
140 Saddle rail
142 Spline bushing
144 Threaded bulkhead
146 Upper threaded coupler mount
148 Gear reduction unit/step-up torque unit
150 Lower motor dampening spacer
152 O-ring
154 Threaded and dual ported coupler
156 Threaded and dual ported lead guide plate
158 Dampening spacer
160 Threaded and dual ported coupler
162 O-ring
164 Lower linkage module
166 Upper linkage module
573 Upper lead guide bearing

PARTS LIST-MECHANIZED EMBODIMENT

11 Upper post
13 Lower post
15 Collar
17 Dust seal wiper
19 Alignment keys
21 Base inflation valve
23 Air chamber
25 O-ring
27 O-ring
29 Threaded bottom cap
31 Controller
33 Piston cavity
35 Valve ball
37 Valve seat
39 Air aperture
41 Threaded orifice coupler insert
43 High helix lead ball screw
45 Piston
47 O-ring
49 O-ring
51 Compression spring
53 Affixed lead ball nut
55 Mid-lead guide bearing
57 Threaded and ported lead guide plate
59 Gear reduction unit
61 Lower gear reduction driver
63 Upper input driver
65 Bevel gear mechanism
67 Splined satellite bevel gear assemblies
69 Rail clamp assemblies
71 Controller housing
73 Hydraulic muffler fitting
75 Body circlip
77 Orifice
79 Inlet manifold
81 Air distribution system
83 Discharge manifold
85 O-ring
87 Piston pathway
89 Body circlip
91 Lower guide bearing
93 Ball nut fasteners
95 Gland bushing
97 Gland O-ring
99 Threaded and ported lead guide plate
101 O-ring
103 Upper guide bearing
105 O-ring
107 Lower threaded gear reduction coupler mount
109 Dampening assembly

111 Lower dampening spacer
113 Reduction gears
115 Upper dampening assembly
117 Exterior rail clamp
119 Hex rail fastener
121 Saddle rail
123 Bulkhead assembly
125 Spline bushing
127 Hex rail fastener
129 Threaded bulkhead
131 Upper threaded coupler mount
133 Upper gear assembly fasteners
135 Tilt adjustment key
137 Lower gear assembly fasteners
139 Body circlip
141 Body circlip
143 Spline bushing
145 Mid-lead guide bearing
147 Lead screw bumper
149 O-ring
151 O-ring
153 Upper post pathway
155 Body circlip
157 Valve ball
159 Hydraulic oil chamber
161 Air inlet bushing

Adjusting Seat Tilt as a Function of Bicycle Inclination

There will now be described further embodiments of the disclosure that relate generally to devices and methods for adjusting the tilt of a bicycle saddle as a function of the inclination or slope of the trail or other route over which the bicycle is moving. The current inclination of the bicycle may be determined, for example, by one or more sensors, such as inertial sensors or GNSS sensors. Based on the trail inclination, a processor may determine an optimum or range of acceptable saddle tilt angles (which may be referred to as a "target saddle tilt angle") that correspond to the determined trail inclination. The processor may then determine the current tilt angle of the saddle. Based on the target saddle angle that corresponds to the determined trail inclination, and based on the current tilt angle of the saddle, the processor determines whether to drive tilting of the saddle. If tilting is needed, the processor instructs a drive unit, such as a motor, to drive tilting of the saddle to the desired target angle. Thus, tilting of the saddle may be performed autonomously, without requiring rider intervention.

The following embodiments may be used in conjunction with any of the above-described devices and methods for adjusting a height of the bicycle saddle. For example, the tilt adjustment devices described below may be used in combination with the motor-driven height adjustment systems described above in connection with FIGS. **1-4**, and in so doing the tilt adjustment devices described below may replace the motor-driven tilting systems described above in connection with FIGS. **1-4**.

Turning to FIG. **10**A, there is shown an embodiment of a seat post assembly **500** according to an embodiment of the disclosure. Seat post assembly **500** comprises a tilt adjustment device **502** mounted on top of a bicycle seat post **510**. Tilt adjustment device **502** comprises a seat post coupling **508** for coupling tilt adjustment device **502** to seat post **510**. Tilt adjustment device **502** further includes a seat coupling comprising a pair of saddle clamps **506** (only one of which can be seen in FIG. **10**A) for coupling a bicycle saddle (not shown) to tilt adjustment device **502**. Tilt adjustment device **502** further includes an exterior housing **504** for housing various components that are used to perform the various functions of tilt adjustment device **502**.

FIG. **10**B shows tilt adjustment device **502** decoupled from seat post **510**. Thus, tilt adjustment device **502** may be retrofitted to an existing bicycle seat post. Alternatively, tilt adjustment device **502** may be provided in combination with seat post **510** for coupling to a seat tube of a bicycle, in which case the seat post **510** may be inserted directly into the seat tube of the bicycle.

Referring to now to FIGS. **11**A-**11**C, there are shown views of an interior of tilt adjustment device **502** with external housing **504** removed for clarity. As can be seen in FIG. **11**B, tilt adjustment device **502** comprises a Printed Circuit Board (PCB) **512** on an upper side of which is provided a power supply such as one or more batteries **511**. PCB **512** further comprises a microprocessor **530** operable to receive and process readings obtained by an inertial sensor module **532** (described below) among other components. Further connected to PCB **512** is a memory card slot **531** for receiving a memory card that may be read by microprocessor **530**.

Turning to FIG. **11**C which shows an underside of PCB **512**, inertial sensor module **532** and a magnetic tilt sensor **533** are connected to PCB **512**. Inertial sensor module **532** comprises a three-axis gyroscope and a three-axis accelerometer. The accelerometer and gyroscope are operable to obtain readings of, respectively, an angular velocity and an acceleration of tilt adjustment device **502**. Magnetic tilt sensor **533** (e.g. such as a **3**D magnetic flux sensor) is operable to generate readings of magnetic flux. Such readings may be used by microprocessor **530** to determine a current angle of saddle clamps **506** relative to seat post coupling **508**, as described in further detail below.

At a rear of tilt adjustment device **502** is provided a stepper motor **524** operable to drive tilting of the bicycle saddle. In particular, in response to one or more instructions received from microprocessor **530**, stepper motor **524** may drive tilting of the saddle by driving rotation of saddle clamps **506** through rotation of a driving member or linkage **522**, described in further detail below. Adjacent to stepper motor **524**, there is provided a further PCB **526** to which is connected a GNSS device **525** (such as a Global Positioning System (GPS) sensor or device).

At a bottom of tilt adjustment device **502** there is shown an upper portion of seat post **510** comprising threads **518** at an end thereof. Threads **518** are configured to couple seat post **520** to seat post coupling **508** of tilt adjustment device **502**.

Turning to FIGS. **12**A and **12**B, there are shown further views of the interior of tilt adjustment device **502**, with certain components removed for clarity. Saddle clamps **506** are connected to a linkage assembly comprising a pair of first linkages **522***a* and **522***b* which in turn are pivotally connected to a second linkage **534** which itself is pivotally connected to a connecting member **514**. Stepper motor **524** is connected to and operable to drive translation of a shaft **527** interconnected with connecting member **514**. Linkages **522***a* and **522***b* define a space therebetween through which shaft **527** extends and through which shaft **527** translates during operation of shaft **527**. As shaft **527** is translated linearly between linkages **522***a* and **522***b* by stepper motor **524**, linkage **534** is pivotally rotated relative to connecting member **514** about a pivot point **535**. Rotation of linkage **534** drives rotation of linkages **522***a* and **522***b* about a pivot point **523**, which in turn drives rotation of saddle clamps **506** about an axis of rotation **537**, thus rotating the bicycle saddle.

FIG. 12C shows linkage 522 with a magnet 539 provided thereon. Magnet 539 is offset from axis of rotation 537 by a first offset, and is offset from magnetic tilt sensor 533 by a second offset. Depending on the angle of rotation of linkage 522 relative to magnetic tilt sensor 533, the distance between magnet 539 and magnetic tilt sensor 533 will vary, and as a result the magnetic flux detected by magnetic tilt sensor 533 will also vary. Depending on the magnetic flux detected by magnetic tilt sensor 533, and based on the first and second offsets, microprocessor 530 is able to determine the particular angular rotation of linkage 522 relative to magnetic tilt sensor 533, and hence the particular tilt angle of the bicycle saddle.

Turning to FIG. 13, there is shown a flow diagram showing an example method 600 of determining a current inclination of the bicycle. FIG. 17 shows an inclination angle of a bicycle 900 relative to a reference position, which in the case of FIG. 17 is a horizontal axis 950. Inclination of the bicycle is therefore synonymous with inclination or slope of the route or trail over which the bicycle is moving.

As described above, depending on the particular inclination of the bicycle, there exist one or more optimum or preferred tilt angles for the bicycle saddle. For example, when riding downhill, it may be preferable to tilt the saddle upwardly from the horizon. Conversely, when riding uphill, it may be preferable to tilt the saddle downwardly from the horizon. Providing a more optimal tilt angle for the saddle depending on when the bicycle is moving uphill or downhill (and therefore depending on the particular inclination of the bicycle) may result in more efficient and/or comfortable riding.

Returning to FIG. 13, at block 602, one or more readings are obtained from the accelerometer of inertial sensor module 532, and at block 604 one or more readings are obtained from the gyroscope of sensor module 532. According to some embodiments, the gyroscope readings may be obtained before the accelerometer readings, or in parallel to the accelerometer readings. At block 606, microprocessor 530 processes the readings. For example, microprocessor 530 may apply one or more suitable digital processing methods or algorithms for processing the readings, such as for example by applying one or more filters. At block 608, microprocessor 530 determines an estimate of the current inclination of the bicycle, based on the processed readings.

In parallel to the method shown in FIG. 13, microprocessor 530 performs a further process as illustrated in FIG. 14. FIG. 14 shows a flow diagram of a method 700 of adjusting a tilt of a bicycle saddle as a function of bicycle inclination. At block 702, microprocessor 530 determines the current inclination of the bicycle (as per, for example, the method shown in FIG. 13). At block 704, microprocessor 530 determines a target tilt angle, based on the determined inclination of the bicycle. For example, microprocessor 530 may refer to a look-up table or similar database stored in a computer-readable memory comprised in tilt adjustment device 502. The look-up table may comprise a list of optimum or preferred saddle tilt angles as a function of bicycle inclination. The contents of the look-up table may be user-configurable. For example, depending on the particular rider, certain riders may prefer to have their saddle tilted to certain predetermined tilt angles depending on the bicycle's particular inclination. Such preferences may be encoded in the look-up table.

Having determined the target tilt angle of the bicycle saddle, at block 706, microprocessor 530 determines a current tilt angle of the bicycle saddle. In particular, magnetic tilt sensor 533 obtains one or more readings of magnetic flux. As described above, the particular angular rotation of linkage 522 relative to magnetic tilt sensor 533 will affect the magnetic flux detected by magnetic tilt sensor 533. The readings from magnetic sensor 533 are received and processed at microprocessor 530. The magnetic flux readings enable microprocessor 530 to determine the particular angular position of linkage 522 relative to magnetic tilt sensor 533. Microprocessor 530 may then determine the current tilt angle of the bicycle saddle. FIG. 16 shows an example of a tilt angle of a bicycle saddle 900 relative to a horizontal reference 902.

By comparing the current tilt angle of the bicycle saddle to the target tilt angle as determined from the look-up table, at block 708, microprocessor 530 determines whether to drive tilting of the saddle. For example, if the current tilt angle of the saddle is within a predetermined threshold (such as a hysteresis band threshold) of the target tilt angle of the saddle, microcontroller 530 may determine that the saddle is already at an acceptable position and that no tilting is required. Conversely, if the current tilt angle of the saddle is not within a predetermined threshold (such as a hysteresis band threshold) of the target tilt angle of saddle, microprocessor 530 may determine that the saddle is not at an acceptable position and that tilting is required.

If tilting is required, then, at block 710, microprocessor 530 instructs stepper motor 524 to initiate tilting of the saddle by rotating rail clamps 506 as described above. At block 712, microprocessor 530 determines whether the tilt angle of the saddle is acceptably close to (e.g. within a predetermined threshold of) the target tilt angle. If the tilt angle of the saddle is acceptably close to the target tilt angle, then the process returns to block 702. If the tilt angle of the saddle is not acceptably close to the target tilt angle, then the process returns to block 710.

When the rider is sitting on the saddle, the load exerted on the saddle by the rider's weight is generally too great for stepper motor 524 to drive tilting of the saddle. If on the other hand the rider is off the saddle (for example if the rider is riding in an upright, unseated position, typically when travelling downhill), then there is effectively no load exerted on the saddle and stepper motor 524 may drive tilting of the saddle. Therefore, prior to instructing stepper motor 524 to drive tilting of the saddle, microprocessor 530 may first determine the current load applied to a reference area, such as the saddle or seat post coupling 508. For example, microcontroller 530 may obtain a reading from a weight sensor or similar sensor positioned to determine a load applied to the saddle.

According to a further embodiment of the disclosure, instead of using inertial sensor module 532 to determine the current inclination of the bicycle, the current inclination of the bicycle may be determined based on one or more GNSS readings. For example, referring still to FIG. 14, at block 702, the current inclination of the bicycle may be determined based on a current geographical location or position of the bicycle.

By dynamically adjusting bicycle seat tilt to an optimum or improved position on-the-fly, the rider may benefit from reduced compression on various anatomical parts, such as their intervertebral discs, sit bones, tail bone, and nerves. In particular, pressure on the perennial nerve (and the pubic bone in women) may be reduced. Furthermore, for sustained climbing efforts, appropriate saddle tilt may maintain regular levels of vessel blood flow in both men and women.

Referring now to FIG. 15, there is shown a flow diagram of a method 800 of determining a current inclination of the bicycle based on one or more GNSS readings. At block 802, one or more GNSS readings are obtained by GNSS device **525**. At block **804**, microprocessor **530** determines a current geographical location of the bicycle based on the one or more GNSS readings. For example, microprocessor **530** may access a topological map stored on a computer-readable medium received in memory card slot **531**. The topological map may comprise data relating to trail or route inclination, slope, or steepness as a function of geographical location. For example, the topological map may comprise data relating to a predetermined route's inclination and elevation for a given position or horizontal distance along the route. For instance, the topological map may comprise a map as provided, for example, by the applications Trailforks™ or Strava™. According to some embodiments, the topological map may store data relating to any absolute geographical position's slope or inclination. This may be particular useful if, for example, the bicycle is not following a designated route or trail, but instead is exploring the backcountry and is "off the beaten track".

Having obtained the one or more GNSS readings, and having accessed the topological map, at block **806**, microprocessor **350** determines the current inclination of the bicycle. For example, the one or more GNSS readings may be used by microprocessor **530** to determine a current geographical location of the bicycle, and the microprocessor **350** may identify in the topological map a geographical location that corresponds to the bicycle's current position. Based on the corresponding geographical location in the topological map, at block **808**, microprocessor **350** may determine the current inclination of the bicycle. After having determined the current inclination of the bicycle, this data may then be used to determine whether tilting of the saddle is required, as per method **700** in FIG. **14**.

Any of the various other features described below may be used in conjunction with tilt adjustment device **502**.

For example, according to some embodiments, a controller may be provided on the bicycle (for example on the handlebars) and may be activated by the rider in order to control tilting of the saddle "on the fly", that is while riding. The controller may be connected to microcontroller **530** using any suitable communication medium. For example, the controller may be connected to microcontroller **530** using any suitable wired or wireless means (such as, for example, via Bluetooth™, Zigbee™, Z-Wave™, ANT+, or Wi-Fi). In response to actuation of the controller, microcontroller **530** may initiate tilting of the saddle by sending a corresponding instruction to stepper motor **524**.

The controller may be manually operable. According to some embodiments, instead of a manually operable controller, the controller may be voice or gesture-activated. For example, using a suitable acoustic or optical sensor (such as a camera or radar-based device), the rider may utter a voice command, or may make a suitable gesture, in order to initiate/cease tiling of the saddle. For example, the rider could speak the words "tilt up fifteen", and an acoustic sensor could relay the voice data to microcontroller **530** whereupon microcontroller **530** may, using suitable voice recognition software, instruct titling of the saddle upwardly from the horizon by 15 degrees.

When controlling tilting of the saddle using a controller as described above, autonomous tilting of the saddle as a function of bicycle inclination (as per the method of claim **14**, for example) may be temporarily halted. For example, in response to tilting of the saddle using a controller as described above, microcontroller **530** may, for instance, cease for a predetermined period of time looping of the tiling algorithms described in FIGS. **13-15**.

According to some embodiments, in addition to having a first controller that may be used to control tiling of the saddle, the bicycle may be further provided with a second controller that may be used to control raising and lowering of the saddle relative to the bicycle seat tube. In this case, the mechanism used to drive raising and lowering of the saddle may be, for example, the motor-driven height adjustment system described above in connection with FIGS. **1-4**. The first controller and the second controller may comprise a single controller. In other words, both tiling of the saddle and raising/lowering of the saddle may be controlled independently of one another via a single controller.

While tilt adjustment device **502** has been described as comprising a stepper motor, a magnetic tilt sensor, and an inertial sensor module for use in driving tilting of the saddle, the skilled person will recognize that other types of seat tilt drivers and sensors may be used without departing from the scope of the disclosure. For example, other types of seat tilt drivers that may be used include any other suitable electric motor such as a brushed or brushless DC electric motor. According to some embodiments, other forms of seat tilt drivers, not limited to motors, may be used. Furthermore, other types of sensors that may be used for determining the current tilt angle of the saddle include, for example, a potentiometer or optical sensor.

Furthermore, while tilt adjustment device **502** has been described as comprising a number of interconnected linkages for driving tilting of the saddle, the skilled person will recognize that other forms of mechanical devices may be used to transfer the driving power from the motor to the saddle clamps.

According to some embodiments, saddle tilting may be driven based on future trail inclination, as anticipated by GNSS readings. For example, the microprocessor may determine changes in trail or route steepness/inclination before the bicycle has reached such points, and therefore may proactively drive tilting of the saddle in anticipation of such changes in route steepness/inclination.

Still further, according to some embodiments, the determined inclination of the bicycle, as determined by the inertial sensor module, may be adjusted by one or more readings from the GNSS sensor. Thus, the inclination of the bicycle may be determined based on a combination of readings from the inertial sensor module as well as the GNSS sensor.

According to some embodiments, seat post **510** may be a telescoping seat post, and for example may comprise a lower seat tube operable to translate relative to an upper seat tube that is connected to tilt adjustment device **502**. In such embodiments, seat post **510** may house one or more containers of a pressurized gas (such as canisters of pressurized CO_2). This may enable a rider to regulate the volume of pressurized gas in the lower seat tube and essentially "top up" at any time the pressure within the lower seat tube. Advantageously, with such a pressurized container stored onboard the bicycle, there may be no need for the rider to carry a manual shock pump. In addition, in the event of a flat tire, the rider may access a stored canister and use the pressurized gas to re-inflate the tire after it has been repaired. Advantageously, with such a pressurized container stored onboard the bicycle, there may be no need for the rider to carry a manual tire pump. The one or more containers may be stored at the base of seat post **510** (e.g. within the lower seat tube), out of sight until the rider wishes to access a canister.

Turning to FIG. **18**, according to some embodiments, seat post **510** may be configured with an anti-rotation mechanism such that relative rotation between the lower seat tube **510***a* and the upper seat tube **510***b* comprised in seat post **510** may be prevented. For example, according to some embodiments, one or both of lower seat tube **510***a* and upper seat tube **510***b* may include one or more protrusions (not shown in FIG. **18**) designed to fit into corresponding grooves, channels, or recesses (not shown in FIG. **18**) in one or both of the other of lower seat tube **510***a* and upper seat tube **510***b*. Such a mating between the one or more protrusions and the grooves, channels, or recesses may prevent relative rotation between lower seat tube **510***a* and upper seat tube **510***b*.

In order to decrease potential wobble during translation of lower seat tube **510***a* relative to upper seat tube **510***b*, the respective shapes of lower seat tube **510***a* and upper seat tube **510***b* may form the anti-rotation mechanism. For example, as can be seen in FIG. **18**, lower seat tube **510***a* and upper seat tube **510***b* may have non-circular cross-sections that prevent relative rotation between lower seat tube **510***a* and upper seat tube **510***b*. For instance, lower seat tube **510***a* and upper seat tube **510***b* may have elliptical cross-sections. As can be seen in FIG. **18**, the elliptical cross-section may include a pair of linear portions **1802** connecting opposed semi-circular portions **1804**.

Turning to FIGS. **19**A and **19**B, according to some embodiments, decoupling of bicycle seat post assembly **500** from the bicycle frame **1904** may be prevented by using a clamp **1902**. As can be seen, clamp **1902** includes a clamp lock **1906** comprising a central recessed portion **1908***a* and multiple (in the case of FIGS. **19**A and **19**B, six) recessed lobe portions **1908***b* extending from central recessed portion **1908***a*.

According to some embodiments, upward and downward translation of seat post **510** may be configured to be wirelessly activated. In particular, a controller provided for example on the handlebars (such as controller **31** discussed in connection with FIGS. **7** and **8**) may be configured to wirelessly activate a portion of a valve allowing seat post **510** to be telescoped upwardly or downwardly. For instance, turning to FIG. **20**, there is shown an example of such a wireless configuration.

In FIG. **20**, a valve assembly **2002** includes a motor **2008** operable to actuate a pin **2010** for displacing a valve ball **2004** relative to a spring **2006**. Displacement of valve ball **2004** relative to spring **2006** will cause air to flow into or out of seat post **510** (as described for example in connection with the embodiment of FIGS. **7** and **8**). The controller provided on the bicycle's handlebars may be configured to wirelessly communicate with motor **2008** by using, for example, a suitable radio frequency communication. For instance, motor **2008** may be communicative with an RF communication module (not shown) that may receive an instruction from the controller and cause motor **2008** to be activated/deactivated depending on the actuation of the controller. Valve assembly **2002** illustrated in FIG. **20** may be suitably incorporated in any of the embodiments described herein. For example, valve assembly **2002** may replace the valve assembly shown in FIGS. **7** and **8** if, for example, wireless activation of the valve is desired.

According to some embodiments, in addition to enabling automatic tilting of the saddle, seat post assembly **500** may additionally be configured such that the rider may adjust the forward and aft positions of the saddle relative to the seat post. For example, as can be seen in FIG. **21**, there is shown an embodiment of a tilt adjustment device **2100** comprising an upper housing **2150** translatable relative to a base **2160**. Base **2160** is secured to a seat post coupling **2165** which may be coupled to a bicycle seat post. A motor (not shown) housed within upper housing **2150** is operable to drive translation of upper housing **2150** relative to base **2160** and in turn seat post coupling **2165**, using for example a rack **2155** engaged to a pinion of the motor. According to other embodiments, different means of driving translation of upper housing **2150** relative to base **2160** may be employed. The motor may be communicatively coupled to a controller operable by a rider of the bicycle. According to some embodiments, the controller may communicate with the motor using wireless means (such as via one or more radio frequency channels).

According to some embodiments, GNSS device **525** may be remotely accessed by a user in order to determine the location of the bicycle in the event that the bicycle is lost or stolen, for example. For instance, a user may initiate communication with GNSS device **525** by using a suitable wireless communication protocol that permits the user to access location data stored onboard GNSS device **525** or that is periodically broadcast by GNSS device **525**. Advantageously, GNSS device **525** may be configured for satellite communication, such that even in the absence of cellular service, it may be possible for a user to communicate with GNSS device **525** (via one or more satellites, for example) in order for the user to determine the location of the bicycle. This may be particularly advantageous, for instance, in the event that a rider is injured in the backcountry without means of calling for help with their traditional mobile device.

Whilst the disclosure has been described in connection with specific embodiments, it is to be understood that the disclosure is not limited to these embodiments, and that alterations, modifications, and variations of these embodiments may be carried out by the skilled person without departing from the scope of the disclosure. It is furthermore contemplated that any part of any aspect or embodiment discussed in this specification can be implemented or combined with any part of any other aspect or embodiment discussed in this specification.

The invention claimed is:

1. A device for adjusting a tilt of a bicycle seat, comprising:

a seat post coupling for coupling to a bicycle seat post;

a seat coupling for coupling to a bicycle seat;

a seat tilt driver for rotating the seat coupling relative to the seat post coupling;

one or more first sensors for determining an angle of the seat coupling relative to the seat post coupling;

one or more second sensors for determining an angle of the seat post coupling relative to a reference position; and

one or more processors configured to execute computer program code stored on a computer-readable medium to perform a seat tilt adjustment operation comprising:

determining from the one or more first sensors the angle of the seat coupling relative to the seat post coupling;

determining from the one or more second sensors the angle of the seat post coupling relative to the reference position; and

adjusting, based on the determined angle of the seat coupling relative to the seat post coupling, and based on the determined angle of the seat post coupling relative to the reference position, the tilt of the bicycle seat by actuating the seat tilt driver, wherein the one or more first sensors comprise one or more magnetic flux sensors fixed relative to one of the seat coupling and the seat post coupling.

2. The device of claim **1**, wherein adjusting the tilt of the bicycle seat by actuating the seat tilt driver comprises:

accessing a database of seat tilt positions and corresponding bicycle inclinations;

identifying in the database, using the determined angle of the seat post coupling relative to the reference position, one or more corresponding bicycle inclinations and one or more corresponding seat tilt positions;

comparing the identified one or more seat tilt positions to the determined angle of the seat coupling relative to the seat post coupling; and

adjusting, based on the comparison, the tilt of the bicycle seat by actuating the seat tilt driver.

3. The device of claim **1**, further comprising one or more third sensors for determining a load applied to a reference area, wherein the seat tilt adjustment operation further comprises:

determining from the one or more third sensors the load applied to the reference area; and

prior to adjusting the tilt of the bicycle seat, determining, using the determined load, whether to adjust the tilt of the bicycle seat by actuating the seat tilt driver.

4. The device of claim **1**, wherein the reference position is a horizon.

5. The device of claim **1**, further comprising one or more magnetic components fixed relative to the other of the seat coupling and the seat post coupling, and wherein the one or more magnetic flux sensors are for detecting one or more magnetic fields generated by the one or more magnetic components.

6. The device of claim **1**, wherein the one or more second sensors comprise one or more gyroscopes for determining an angular velocity of the device in each of orthogonal x, y, and z axes, and one or more accelerometers for determining an acceleration of the device in each of the x, y, and z axes.

7. The device of claim **1**, further comprising:

a user input device,

wherein the computer-readable medium further comprises computer program code configured when executed by the one or more processors to cause the one or more processors to perform a further seat tilt adjustment operation comprising:

detecting one or more user inputs received at the user input device; and

adjusting, based on the detected one or more user inputs, the tilt of the bicycle seat by actuating the seat tilt driver.

8. The device of claim **7**, wherein the user input device comprises one or more of:

an acoustic sensor for detecting a voice command;

an optical sensor for detecting a gesture; and

a manually operable control device.

9. The device of claim **1**, further comprising:

an adjustable-height seat post for coupling to the seat post coupling and comprising a seat tube coupling for coupling to a bicycle seat tube;

a seat height driver for adjusting a height of the seat post coupling relative to the seat tube coupling; and

a user input device,

wherein the computer-readable medium further comprises computer program code configured when executed by the one or more processors to cause the one or more processors to perform a seat height adjustment operation comprising:

detecting one or more user inputs received at the user input device; and

adjusting, based on the detected one or more user inputs, the height of the seat post coupling relative to the seat tube coupling by actuating the seat height driver.

10. The device of claim **1**, wherein the seat tilt driver is configured to drive a translation of a driving member coupled to a linkage system connected to the seat coupling such that translation of the driving member drives rotation of the seat coupling.

11. The device of claim **10**, wherein the linkage system comprises a first linkage coupled to the driving member, and a linkage assembly connected to the seat coupling and pivotally coupled about a pivot to the first linkage.

12. The device of claim **11**, wherein the linkage assembly comprises a pair of second linkages, and wherein the driving member is operable to be translated between the pair of second linkages.

13. The device of claim **11**, wherein the linkage assembly comprises a magnetic component for magnetically interacting with the one or more magnetic flux sensors.

14. The device of claim **1**, wherein the one or more first sensors are provided on a printed circuit board (PCB), and wherein the one or more second sensors are provided on the same PCB.

15. A device for adjusting a tilt of a bicycle seat, comprising:

a seat post coupling for coupling to a bicycle seat post;

a seat coupling for coupling to a bicycle seat;

a seat tilt driver for rotating the seat coupling relative to the seat post coupling;

one or more first sensors for determining an angle of the seat coupling relative to the seat post coupling;

one or more second sensors for determining an angle of the seat post coupling relative to a reference position; and

one or more processors configured to execute computer program code stored on a computer-readable medium to perform a seat tilt adjustment operation comprising:

determining from the one or more first sensors the angle of the seat coupling relative to the seat post coupling;

determining from the one or more second sensors the angle of the seat post coupling relative to the reference position; and

adjusting, based on the determined angle of the seat coupling relative to the seat post coupling, and based on the determined angle of the seat post coupling relative to the reference position, the tilt of the bicycle seat by actuating the seat tilt driver,

wherein the seat tilt driver is configured to drive translation of a driving member coupled to a linkage system connected to the seat coupling such that translation of the driving member drives rotation of the seat coupling, and

wherein the linkage system comprises a first linkage coupled to the driving member, and a linkage assembly connected to the seat coupling and pivotally coupled about a pivot to the first linkage.

16. A device for adjusting a tilt of a bicycle seat, comprising:

a seat post coupling for coupling to a bicycle seat post;

a seat coupling for coupling to a bicycle seat;

a seat tilt driver for rotating the seat coupling relative to the seat post coupling;

one or more first sensors for determining an angle of the seat coupling relative to the seat post coupling;

one or more second sensors for determining an angle of the seat post coupling relative to a reference position; and

one or more processors configured to execute computer program code stored on a computer-readable medium to perform a seat tilt adjustment operation comprising:

determining from the one or more first sensors the angle of the seat coupling relative to the seat post coupling;

determining from the one or more second sensors the angle of the seat post coupling relative to the reference position; and

adjusting, based on the determined angle of the seat coupling relative to the seat post coupling, and based on the determined angle of the seat post coupling relative to the reference position, the tilt of the bicycle seat by actuating the seat tilt driver,

wherein the one or more second sensors comprise one or more Global Navigation Satellite System (GNSS) sensors for determining a current position of a bicycle, and wherein determining the angle of the seat post coupling relative to the reference position comprises:

determining from the one or more GNSS sensors a current geographic position of the bicycle; and

determining, based on at least the current geographic position of the bicycle, the angle of the seat post coupling relative to the reference position.

17. The device of claim **16**, wherein determining the angle of the seat post coupling relative to the reference position comprises:

identifying one or more slopes corresponding to the current geographic position of the bicycle; and

determining, based on the one or more slopes, the angle of the seat post coupling relative to the reference position.

18. The device of claim **17**, wherein identifying the one or more slopes comprises:

accessing a database of geographic positions and corresponding slopes; and

identifying in the database, based on the current geographic position of the bicycle, the one or more slopes.

19. The device of claim **17**, wherein the geographic positions comprise positions along one or more predetermined routes.

20. A device for adjusting a seat position of a bicycle seat, the device comprising:

a seat tube coupling configured to couple to a bicycle seat tube;

a seat coupling configured to couple to a bicycle seat;

a seat adjustment mechanism movably coupling the seat tube coupling and the seat coupling and comprising:

a prime mover; and

a tilt actuator operable to adjust a tilt of the seat coupling relative to the seat tube coupling, wherein the tilt actuator comprises a linkage system connected between an output of the prime mover and the seat coupling such that operation of the prime mover operates the linkage system and thereby adjusts the tilt of the seat coupling relative to the seat tube coupling; and

a tilt controller communicative with the seat adjustment mechanism and operable by a rider of a bicycle to actuate the prime mover thereby adjusting the tilt of the seat coupling relative to the seat tube coupling.

21. The device of claim **20**, wherein the prime mover is configured to drive a translation of a driving member coupled to the linkage system such that translation of the driving member drives rotation of the seat coupling.

22. The device of claim **21**, wherein the linkage system comprises:

a first linkage coupled to the driving member; and

a linkage assembly connected to the seat coupling and pivotally coupled about a pivot to the first linkage.

23. The device of claim **22**, wherein the linkage assembly comprises a pair of second linkages, and wherein the driving member is operable to be translated between the pair of second linkages.

24. The device of claim **20**, wherein the tilt controller is remote from the seat adjustment mechanism and operable by the rider of the bicycle when the rider is riding the bicycle.

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